The National Institute of Engineering, Mysuru
Electronics and Communication Engineering

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Major project presentation on

"Wireless Sensor Network for Data Extraction"

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contents

Chapter 1

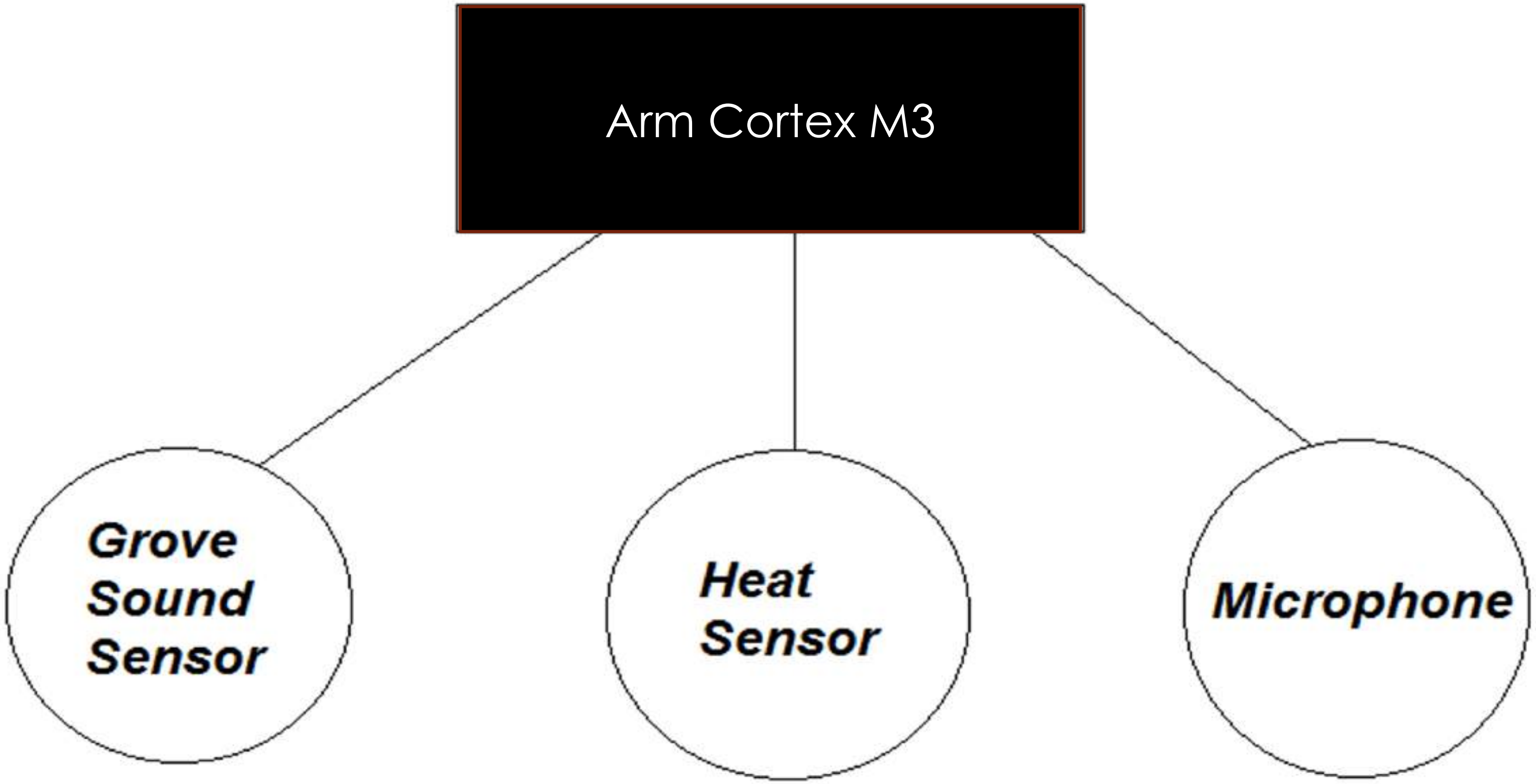
Introduction and Literature Survey

Introduction

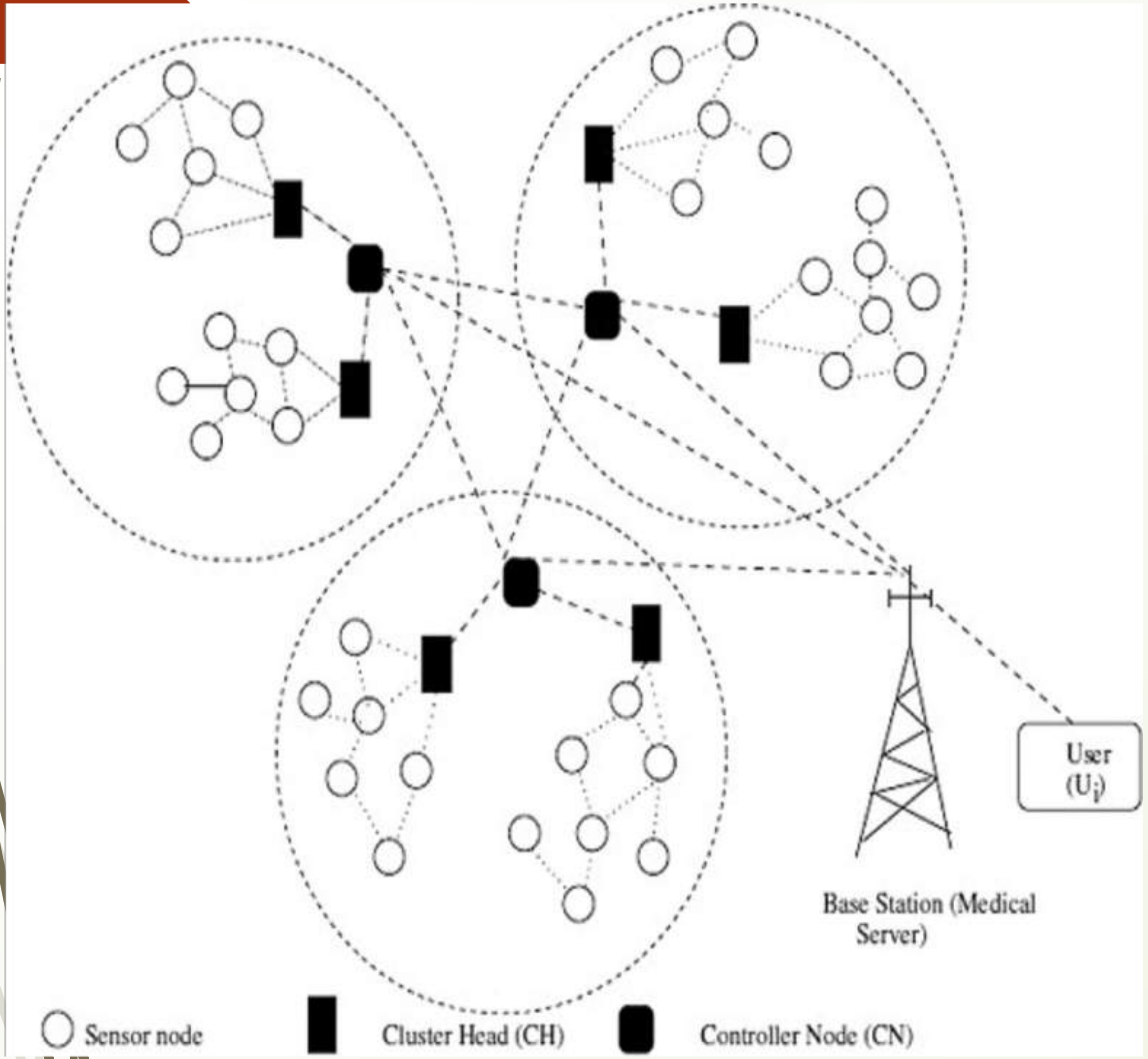
- A wireless network is a communication network made up of radio nodes organized in a specific topology. Nodes can be built with single radio, dual radios, and multiple radios. The number of radios in a node depends on the throughput and latency demands of the wireless network.

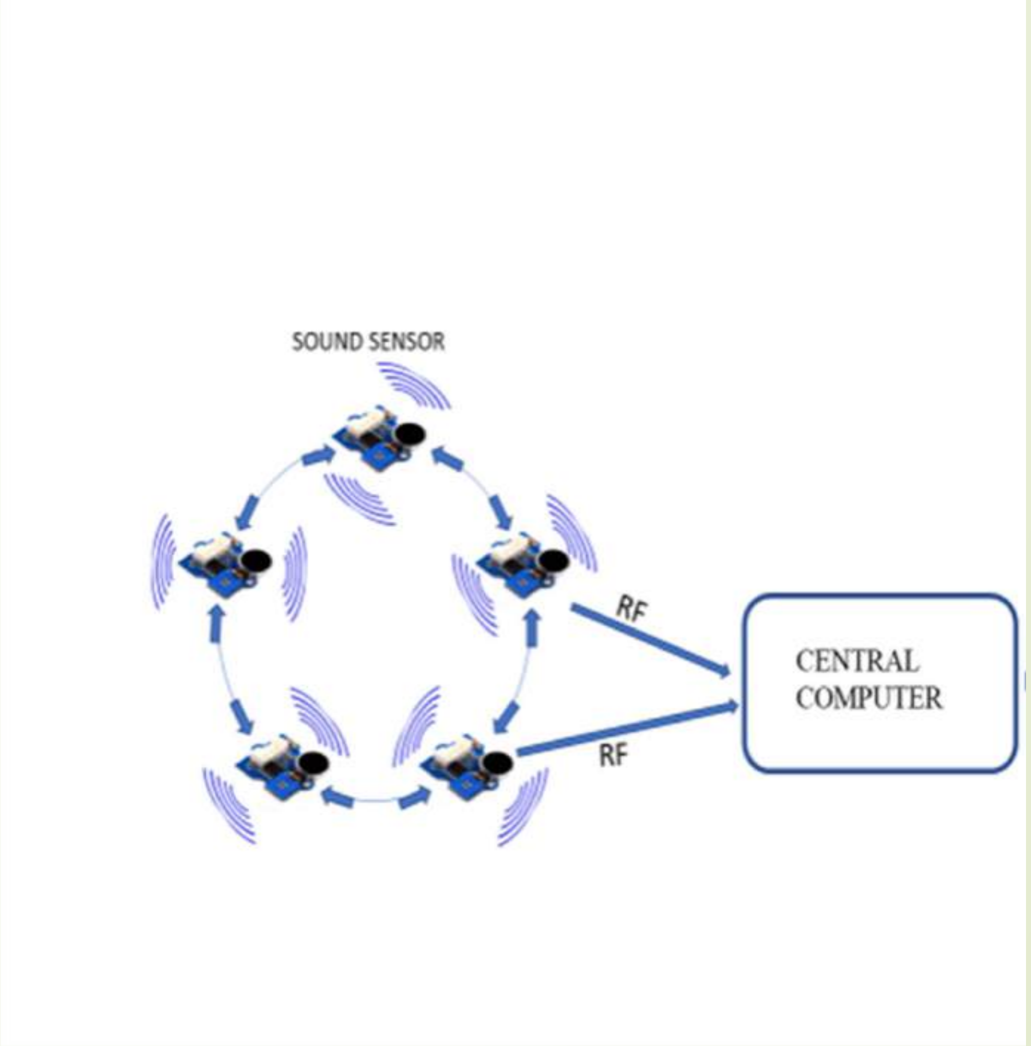
- Modern technological advancements in sensor technology, wireless networking, the decry in device sizes and signal processing have facilitated the design and proliferation of wireless sensor network

- The sensor data is sent to central server from various nodes in a wireless network for further processing. for example the food temperature and humidity in various containers, or multiple camera footages sent to a central server for detection and further processing. Similarly forest fire and forest logging detection can be done using this strategy.
- The response of the central server is transmitted to the required destination thereby avoiding and alerting the danger
- Wireless sensor networks comprise of different types of sensors, that could be temperature sensors, pressure sensors, sound sensors etc., and beam the output wirelessly, either through radio or through the internet.



Components of a node


Base Station (Medical Server)
User (U$_i$)
Sensor node
Cluster Head (CH)
Controller Node (CN)


SOUND SENSOR
RF
RF
CENTRAL COMPUTER

Literature Survey

Author	Title	Advantages	Disadvantages
Ru Chen, Chenggang Zhu, Bilin Ge, Xiangdong Zhu, Yung-Shin Sun, Lan Mi , Jiong Ma, Xu Wang, and Yiyan Fei	Detecting Signals With Direct Fast Fourier Transform for Microarray Data Collection, IEEE Photonics Technology Letters, VOL. 29, NO. 24, December 15, 2017	➢to detect a small periodic signal on top of a large DC background ➢ to amplify signals as small as nV to the range of V with low noise amplifiers before Fourier analysis	➢Power consumption of each node increased. ➢Poor efficiency in data gathering
Naima EL Haoudar ,Prof. Dr. Abdelilah Maach, Department of science and computer Mohammadia School of Engineering	Routing Metric for Wireless Mesh Networks, 978-1-4673-2679-7/12 ©2012 IEEE.	➢Mesh routers form the backbone and they have minimal mobility which guarantees high connectivity ➢High robustness	➢ more sophisticated algorithms and design principles are required for the realization of WMNs.
Stefan Bouckaert, Eli De Poorter, Pieter De Mil, Ingrid Moerman, Piet Demeester	Interconnecting Wireless Sensor and Wireless Mesh Networks: Challenges and Strategies, 978-1-4244-4148-8/09©2009	➢adaptive network protocols are able to adjust their operation characteristics to the individual node needs, while guaranteeing interoperability. ➢ self-organizing and easy	➢Poor signals when affected by noise ➢Interconnecting techniques is lacking

Gokay Saldamli, Sumedh Deshpande, Kaustubh Jawalekar, Pritam Gholap, Loai Tawalbeh , Levent Ertau	"Wildfire Detection using Wireless Mesh Network", 2019 Fourth International Conference on Fog and Mobile Edge Computing (FMEC), 978-1-7281-1796-6©2019 IEEE	To detect wildfires, created sensor nodes connected to ARM Cortex M3 controller, deployed on high trees with a communication interface integrated with it. Implemented a LORA mesh network to beam the sensor data to a central Raspberry Pi, then to a Dashboard on a webpage.	Communication interface integrated with the node need to be more efficient
József Kopják, Gergely Sebestyén	Comparison of data collecting methods in wireless mesh sensor networks, SAMI 2018 • IEEE 16th World Symposium on Applied Machine Intelligence and Informatics February 7-10, Košice, Herl'any, Slovakia	➢ wireless devices cannot cover the whole area of the facility. ➢ With routing the RF range can be extended. ➢ RF network based IQRF technology which uses less power	➢ The data collecting duration is longer ➢ Poor SNR Ratio for longer distance

Chapter 2

Problem formulation and Methodology

- Data from each node sent to a central processor using different techniques which can be through internet, radio transmission or any other methods. A wireless network, where all the nodes are connected to central processor is used.
- Each Sensor is connected with a radio transmitter that transmits the data in a specific frequency. Each sensor shall broadcast the data with its specific ID number such that it is easily located later on. The data is received by the central computer that analyses the signal transmitted using Fourier transform. If a match is found, the ID number of the sensor is used to locate the region of alert.

This TS5828 Mini 48CH 600MW Transmitter is the advanced version of our TS5828 5.8GHz 32CH transmitter. It features widely used 5.8GHz power for transmission.

The all-new TS5828 Mini 48CH 600MW Transmitter packs a whopping 600mW of ultra-clean 5.8GHz power! A must for those looking for longer range and locked in the video.

The TS5828 is transmitting a full range of 48 channels and uses an easy to use interface, it also comes with a clean pre-wired harness, and only weighs in at a scant 15grams, it's Perfect for a wide range of aircraft!

The TS5828 packs 600mw of 5.8GHz power and 48 channels into a micro-sized audio/video transmitter. It weighs just 15g completed with the wire harness and stock antenna, making it perfect for a wide range of aircraft and particularly for smaller 150-250 sized multi-rotors such as

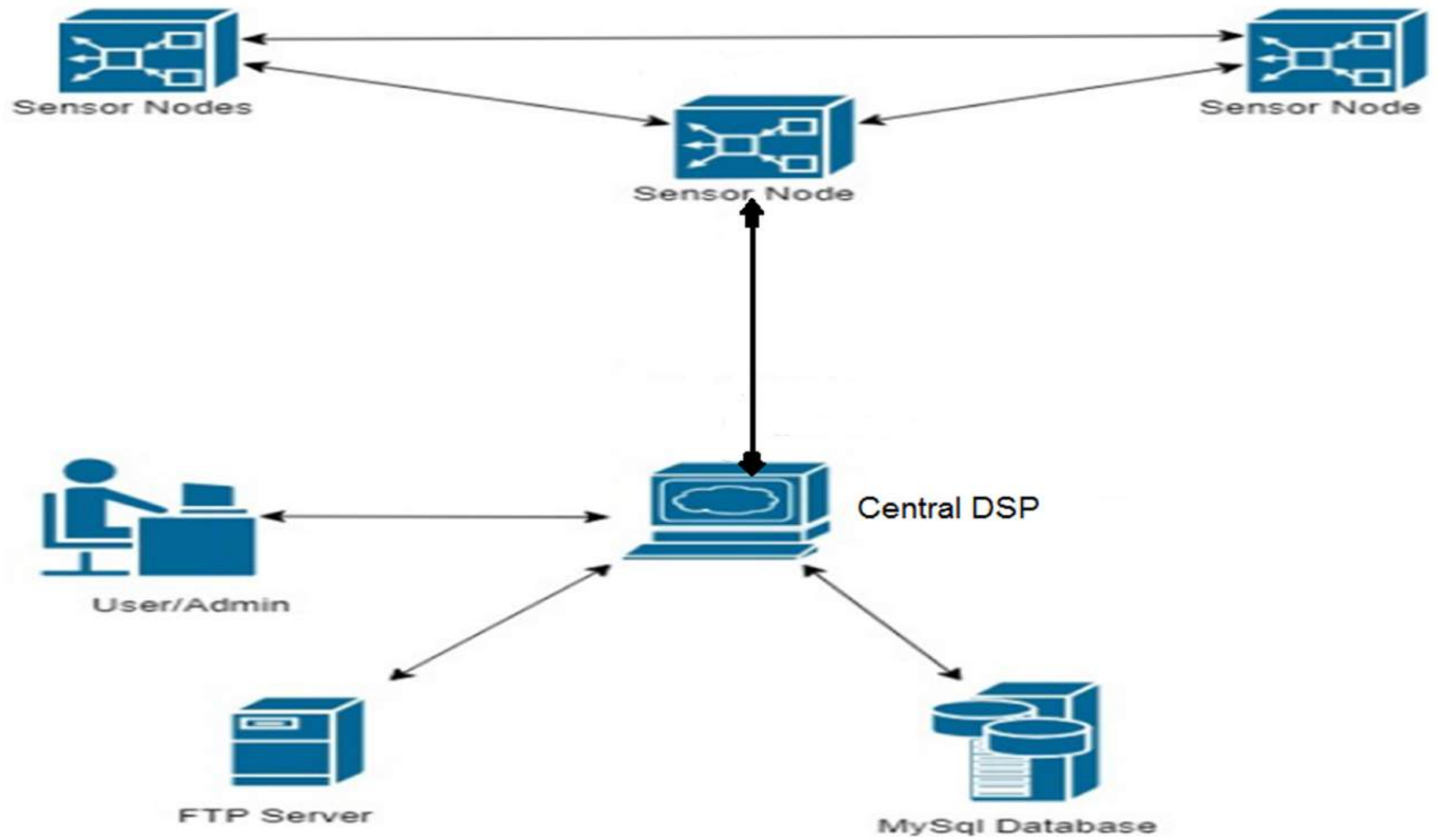
Features :

1. Compact and lightweight
2. Perfect for use with small models
3. Covers A, B, C, D, H, L bands
4. 48 Channels
5. 600mw RF output
6. Wide input voltage range
7. Long range:1~2km with the stock antenna
8. Stable signal

TS5828 Mini 48CH 600MW Transmitter

- We implement the sensors in a artificial honeycomb structure each placed half miles apart, all connected wirelessly to a central office through radio waves. When logging machines enter the assigned area the frequency of the sound produced by the machines is detected by the sound sensors using signal processing techniques and ultimately an alert message is sent to the nearest office of concern.
- The sound sensors are programmed such that they send the information only if they receive a sound higher than 60db.
- The Central Computer is fed with different data sets of the different type of signals that are created by different logging and mining machines. Whenever a match is found by the signal sent by sensors, it alerts the central office. Here, signal processing is used to extract the signal and compare it with the already fed signal for matching.

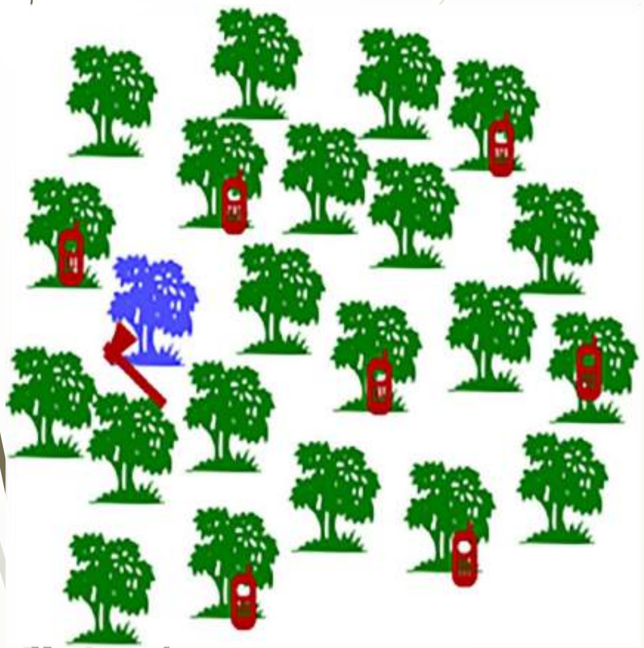
PROPOSED METHOD

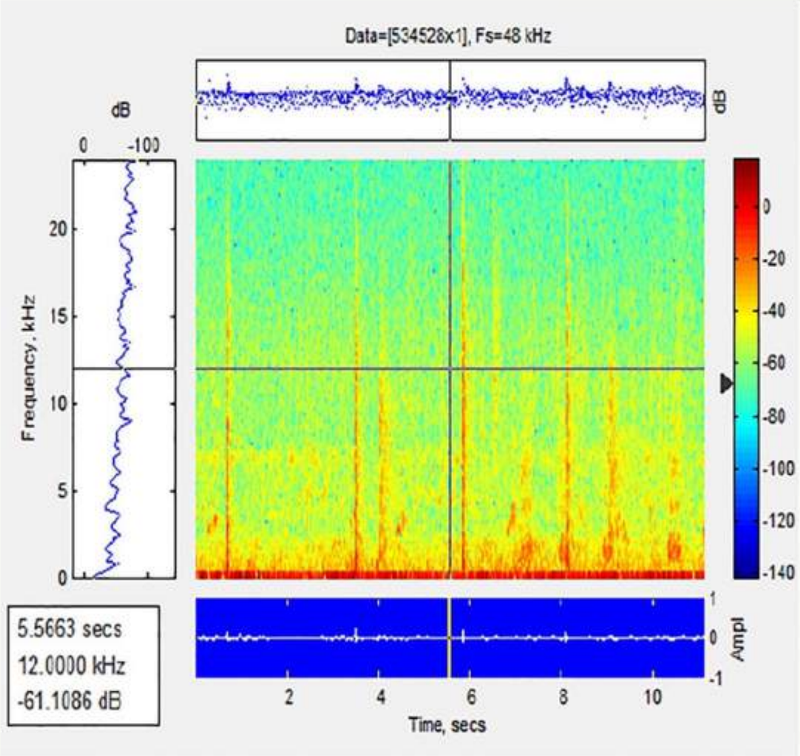


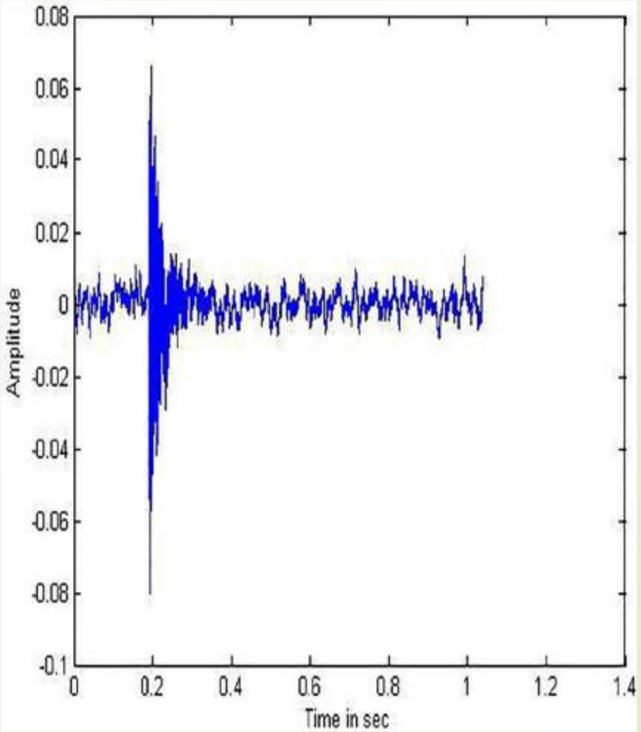
Node setup on a tree

Waveform of single strike by axe on green wood.

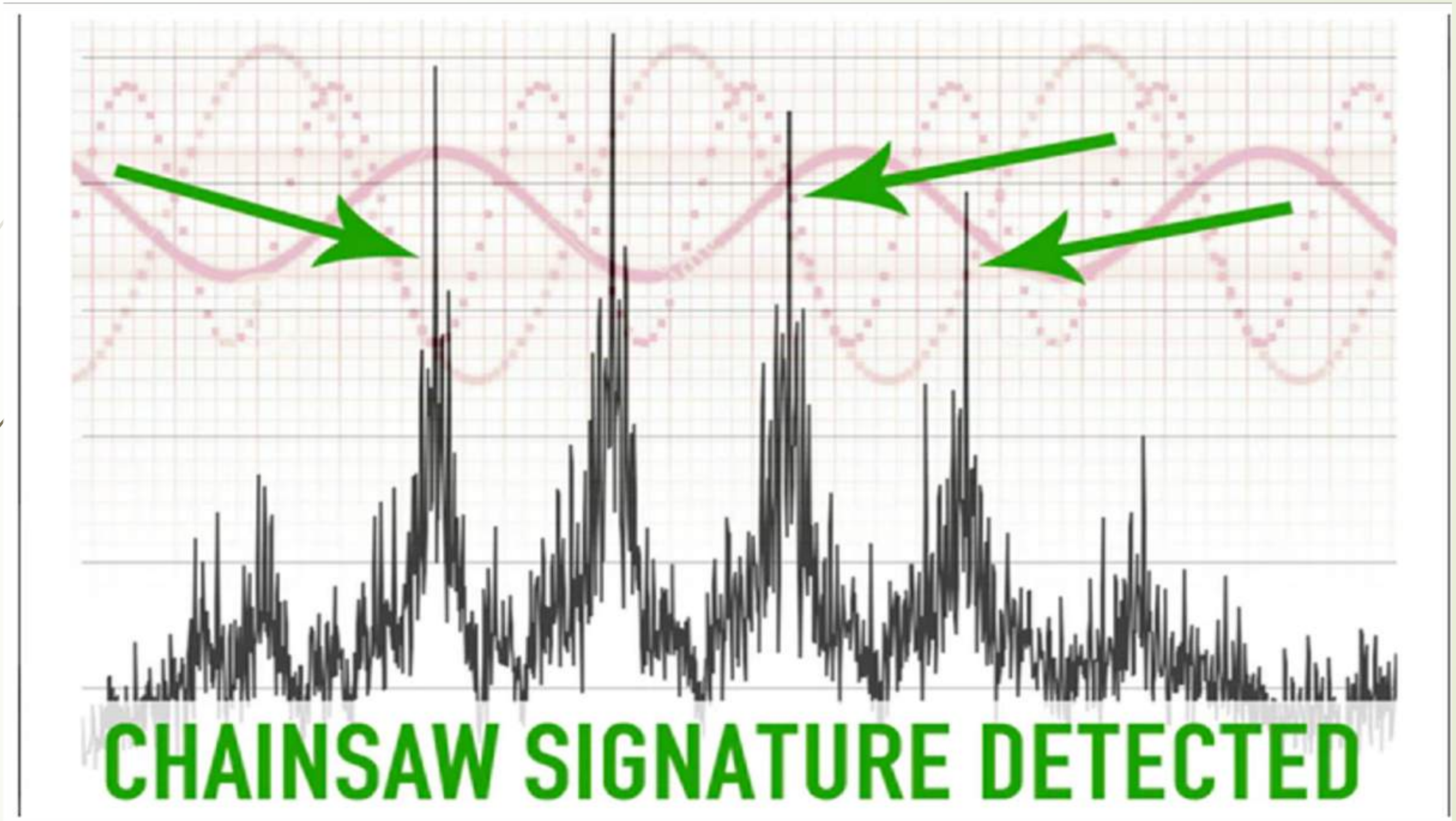
Spectrogram of single strike by axe on green wood.



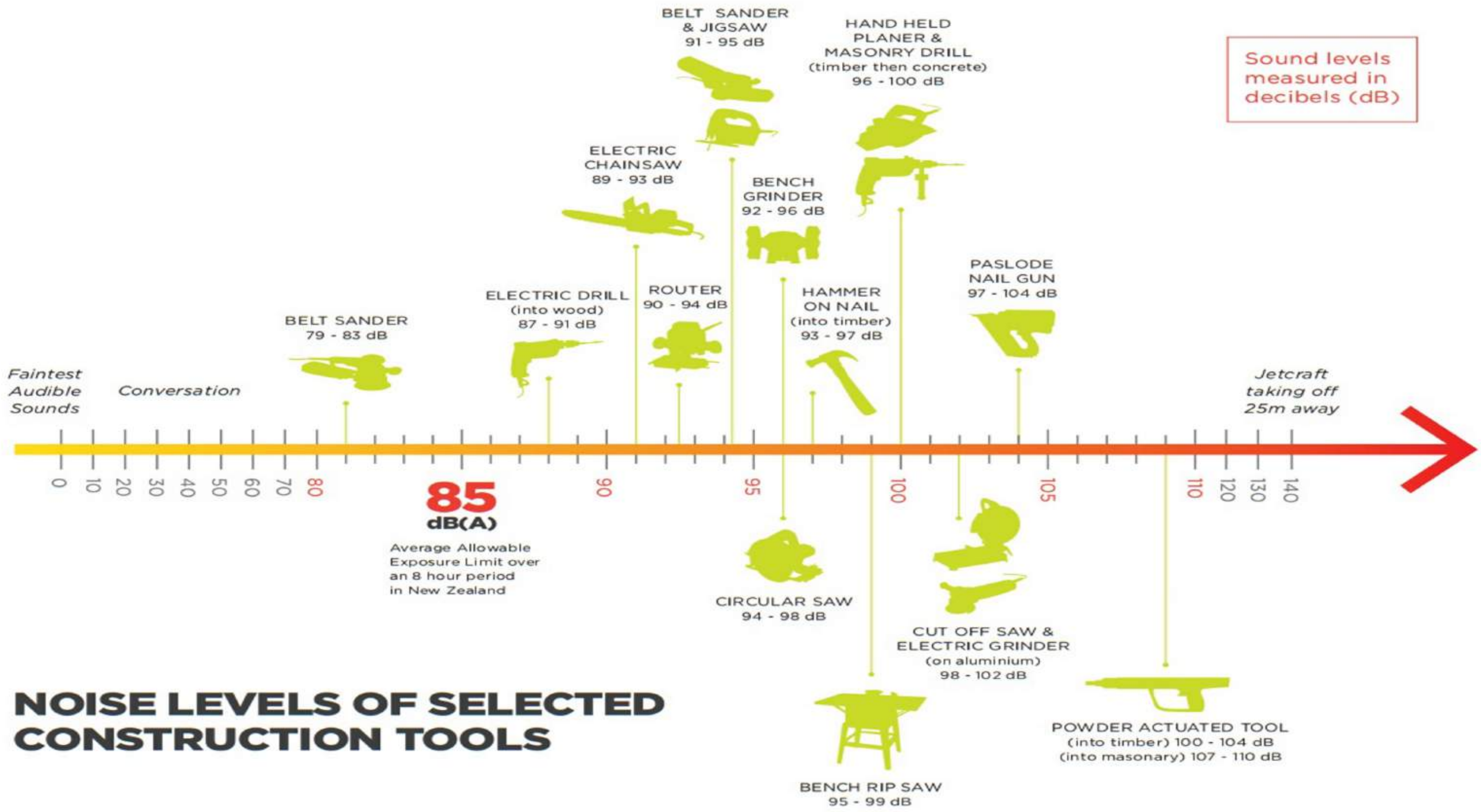




Detection of Signal



Reference -https://www.kickstarter.com/projects/topherwhite/rainforest-connection-phones-turned-to-forest-guar



NOISE LEVELS OF SELECTED CONSTRUCTION TOOLS
Sound levels measured in decibels (dB)
Faintest Audible Sounds
Conversation
BELT SANDER 79 - 83 dB
ELECTRIC DRILL (into wood) 87 - 91 dB
ELECTRIC CHAINSAW 89 - 93 dB
ROUTER 90 - 94 dB
BELT SANDER & JIGSAW 91 - 95 dB
BENCH GRINDER 92 - 96 dB
HAMMER ON NAIL (into timber) 93 - 97 dB
HAND HELD PLANER & MASONRY DRILL (timber then concrete) 96 - 100 dB
PASLODE NAIL GUN 97 - 104 dB
Jetcraft taking off 25m away
0
10
20
30
40
50
60
70
80
85
dB(A)
90
95
100
105
110
120
130
140
Average Allowable Exposure Limit over an 8 hour period in New Zealand
CIRCULAR SAW 94 - 98 dB
BENCH RIP SAW 95 - 99 dB
CUT OFF SAW & ELECTRIC GRINDER (on aluminium) 98 - 102 dB
POWDER ACTUATED TOOL (into timber) 100 - 104 dB (into masonary) 107 - 110 dB

Logging Machines Noise Levels

Machine	Idle (dB)	Full throttle (dB)
Skidder 1997 Franklin Tree Farmer 170 (enclosed cab)	73	100
Skidder 1997 Caterpillar 515 (enclosed cab)	72	84
Skidder 1995 Caterpillar 518C	82	94
Skidder 1964 Franklin Tree Farmer C6	78	102
Skidder 1960 Franklin Tree Farmer C5	82	100
Cutter 1998 Tigercat 845	74	90 (not cutting)
Cutter 1996 Barko 885	76	96 (not cutting)
Loader 1998 Tigercat 860S	68	74
Loader 1998 Tigercat 860S with fan on		82
Loader 1998 Tigercat 860S with fan and radio on		90
Loader 1996 Barko 169B	78	92
Loader 1996 Prentice 210E	80	90
Loader 1960 Barko 160	90	108
Bulldozer 1997 Caterpillar D4H XL	98	102
Bulldozer 1976 John Deere 450 bulldozer	85	98 (¾ throttle)
Bulldozer 1964 Caterpillar D5	84	112
Chainsaw 1998 Stihl 044 (measured 10 feet away)	78	106
Chainsaw 1997 Husqvarna 272 (measured 10 feet away)	76	102
Chainsaw 1995 Husqvarna 268 (measured 10 feet away)	82	104

Software and Hardware

Hardware:

- Generic STM32F103RBT6 ARM STM32 Minimum system development board is used at the nodes in a wireless network and in interfacing the sensors and transmitter module for data transmission and reception.
- Sensors like (MQ-2 MQ2 Smoke Gas LPG Butane Hydrogen Gas Sensor) Detector Module, High Definition 1200TVL CMOS Camera, INMP401 ADMP401 MEMS Microphone Breakout Spark fun,(KG496 Xcluma Sound sensor) are used for analyzing the environment with precision to relay the data to the arm and send audio clips to the central server for classification of the sound.
- Processor-DSP(Tms 6713) to implement signal processing techniques at the central server for distinguishing the sound from audio sensors.

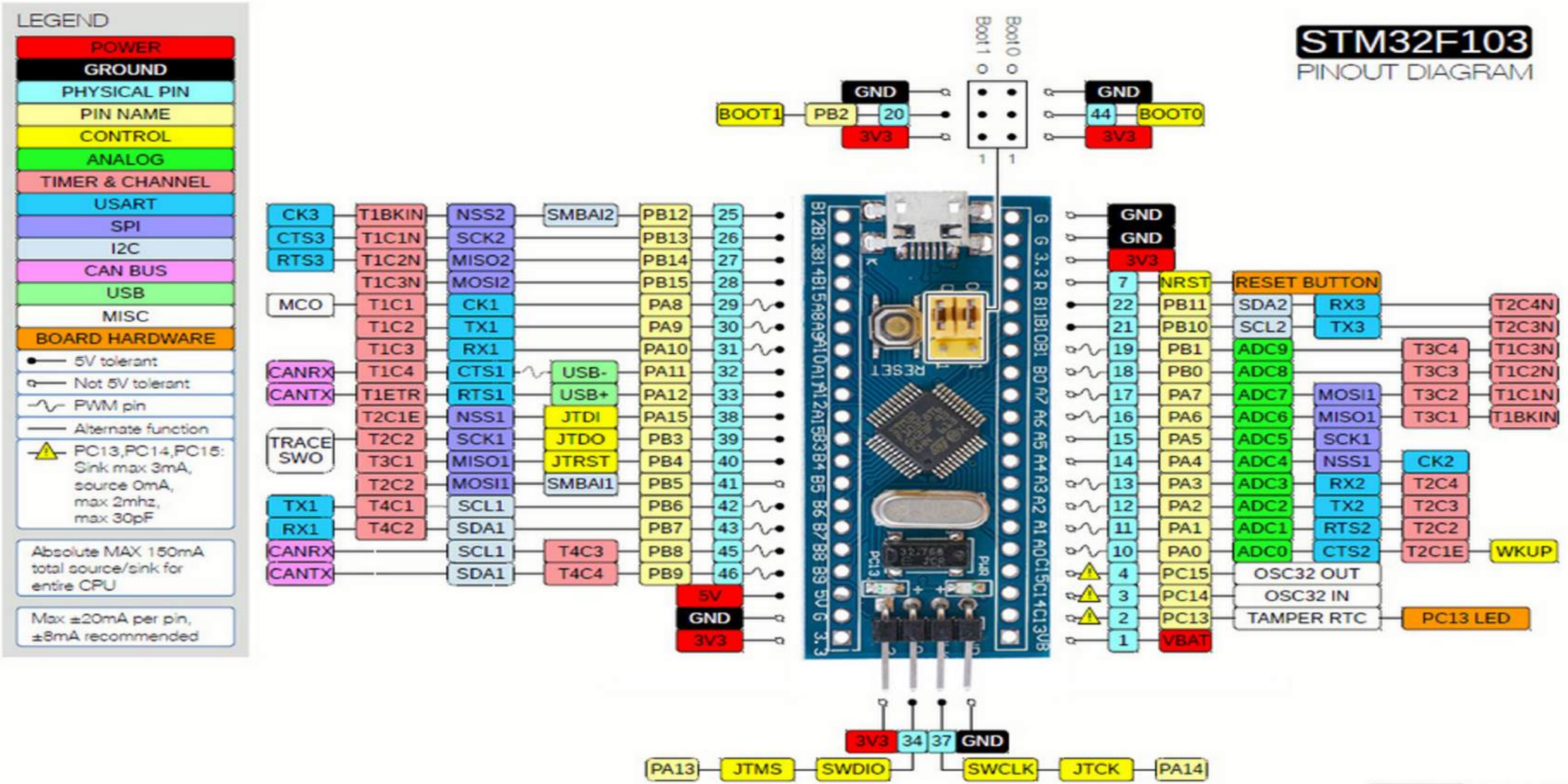
Software:

- MATLAB used in the central server for data classification with signal processing algorithms
- Code composer studio is used to write C codes of algorithms

Chapter 3

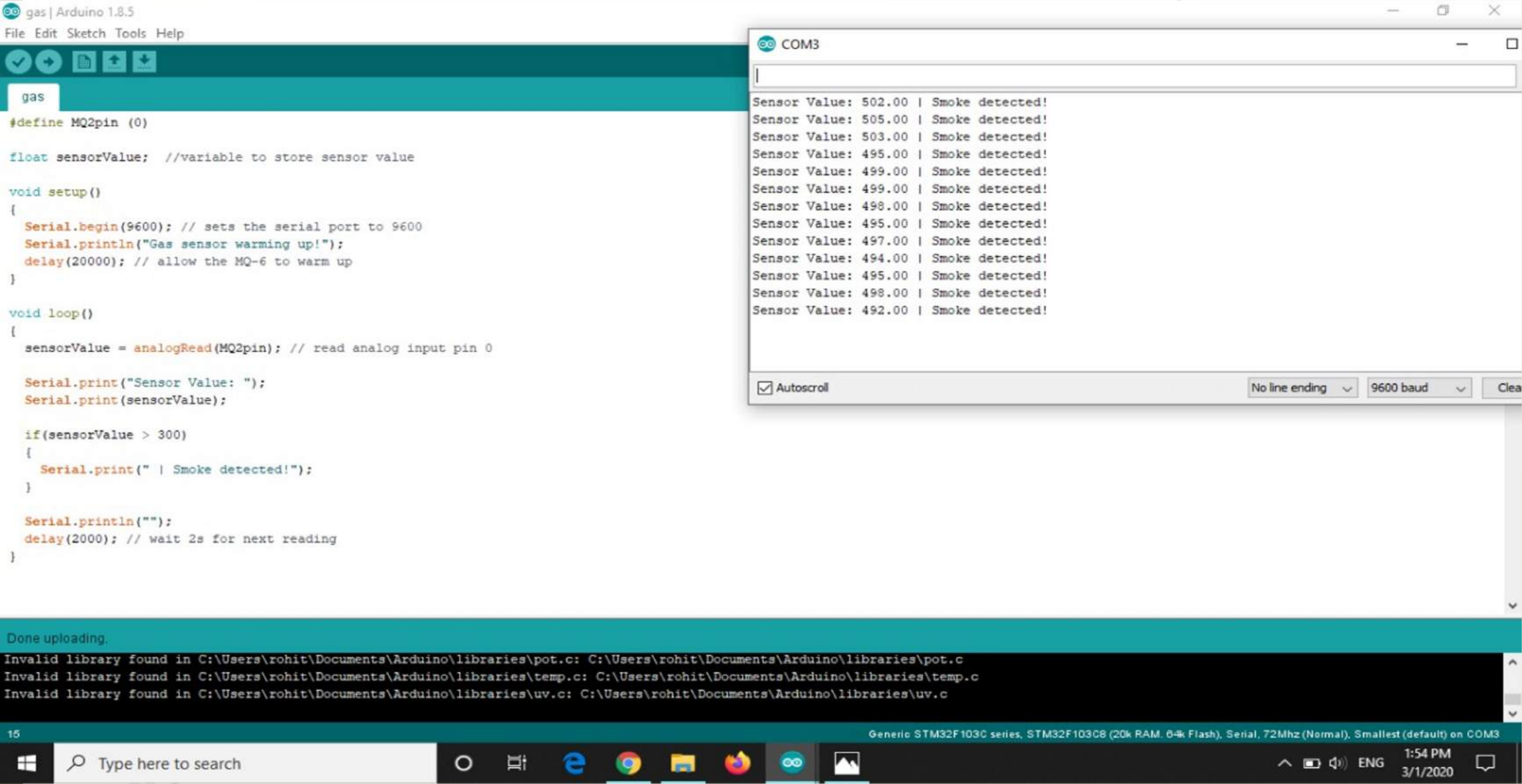
System Design

STM32F103RBT6 PIN DIAGRAM



Node and its sensors

Node Interface with MQ-2 on Arduino platform



Block diagram of DSP processor kit

The C6713 DSK is a low-cost standalone development platform that enables users to evaluate and develop applications for the TI C67xx DSP family. The DSK also serves as a hardware reference design for the TMS320C6713 DSP. Schematics, logic equations and application notes are available to ease hardware development and reduce time to market.

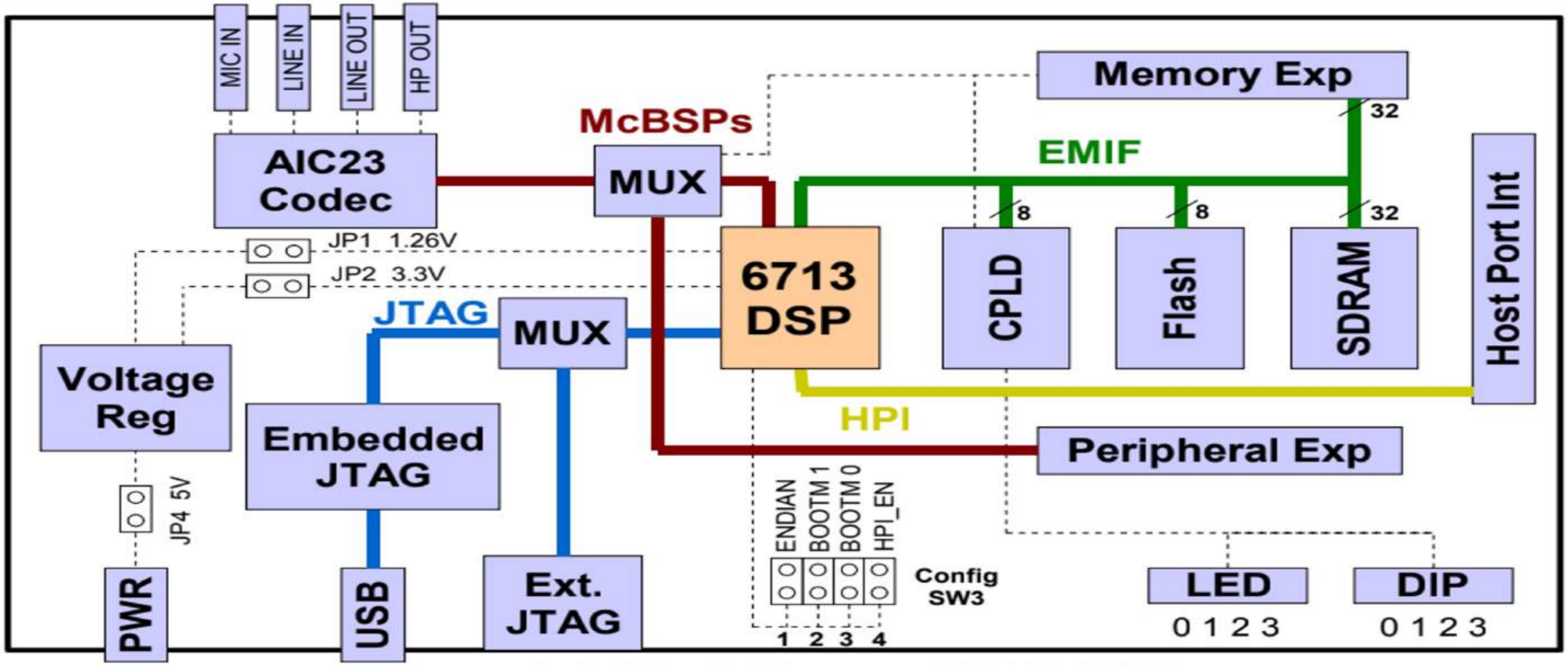


Figure 1-1, Block Diagram C6713 DSK

DSK CODEC INTERFACE

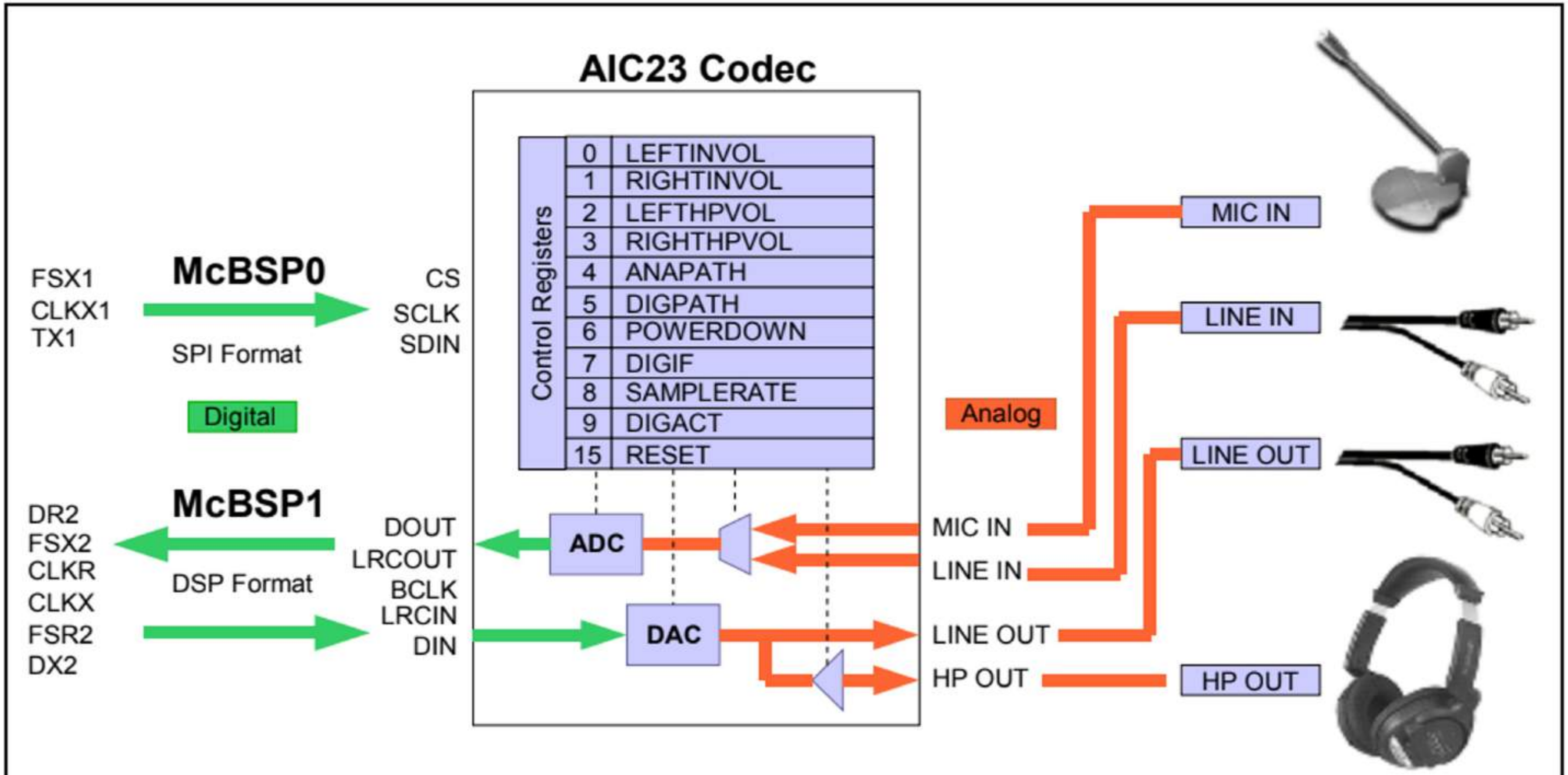


Figure 2-1, TMS320C6713 DSK CODEC INTERFACE

Signal statistical parameters

1. Energy

the energy of a signal is defined using the square of the signal magnitude, the envelope of squared signal magnitude or the integral of squared signal magnitude.

energy of a discrete-time signal x(*n*) is defined mathematically as

$$E_s = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

2. Arithmetic Mean

For a data set, the arithmetic mean, also called the expected value or average, is the central value of a discrete set of numbers: specifically, the sum of the values divided by the number of values. The arithmetic means of a set of numbers $x_1, x_2, \ldots, x_n$ is typically denoted by x bar. If the data set were based on a series of observations obtained by sampling from a statistical population. The *arithmetic mean* (or simply *mean*) of a sample $x_1,x_2,\ldots,x_n$ x 1 , x 2 , … , x n, usually denoted b x bar, is the sum of the sampled values divided by the number of items in the sample.

$$Avg(x) = \frac{1}{n} * \sum_{i=1}^{n} x_i$$

2.2 Zero Crossing Rate

zero crossing rate of any signal frame is the rate at which a signal changes its sign during the frame. It denotes the number of times the signal changes value, from positive to negative and vice versa, divided by the total length of the frame.

Zero crossing rate is computed as:

1.Let $x_i n = 0, 1, N-1$ be the samples of the i^{th} frame.
2.Zero crossing rate of each frame is calculated as:

$$Z = \frac{1}{2 * N} * \sum_{n=0}^{N-1} (sgnx_i n) - (sgnx_i n - 1)$$

Where sgnx$_i$ n is signum function $\begin{cases} 1 & x_i n \geq 0 \\ -1 & x_i n < 0 \end{cases}$

ZCR can be interpreted as a measure of the noiseless of a signal. It usually exhibits higher values in the case of noisy signals
Zero crossings are a basic property of an audio signal that is often employed in audio classification. Zero crossings allow for a rough estimation of dominant frequency and the spectral centroid. It is one of the cheapest and simplest features is the ZCR, which is defined as the number of zero crossings in the temporal domain within one second. ZCR is a popular feature for speech/music discrimination due to its simplicity. However, it is extensively used in a wide range of other audio application domains, such as musical genre classification, highlight detection, speech analysis, singing voice detection in music, and environmental sound recognition.

4. Standard Deviation

In statistics the **standard deviation** is a measure of the amount of variation or dispersion of a set of values. A low standard deviation indicates that the values tend to be close to the mean (also called the expected value) of the set, while a high standard deviation indicates that the values are spread out over a wider range.

The standard deviation of a random variable, statistical population, data set, or probability distribution is the square root of its variance. It is algebraically simpler, though in practice less robust, than the average absolute deviation. A useful property of the standard deviation is that, unlike the variance, it is expressed in the same units as the data.

$$SD = \sqrt{\frac{1}{N-1} * \sum_{i=1}^{N} (x_i - \mu)^2}$$

Where, N= number of samples
x_i=current score or data set
μ=arithmetic mean

5. Range

In statistics, the range of a set of data is the difference between the largest and smallest values. Difference here is specific, the range of a set of data is the result of subtracting the smallest value from largest value.

However, in descriptive statistics, this concept of range has a more complex meaning. The range is the size of the smallest interval (statistics) which contains all the data and provides an indication of statistical dispersion. It is measured in the same units as the data. Since it only depends on two of the observations, it is most useful in representing the dispersion of small data sets.

$$Range = \max imum(x_i) - \text{minimum}(x_i)$$

Where,

X_i=data set

6. coefficient of variation

In probability theory and statistics, the coefficient of variation (CV), also known as relative standard deviation (RSD), is a standardized measure of dispersion of a probability distribution or frequency distribution. It is often expressed as a percentage, and is defined as the ratio of the standard deviation σ to mean (or its absolute value, μ.

It shows the extent of variability in relation to the mean of the population. The coefficient of variation should be computed only for data measured on a ratio scale, that is, scales that have a meaningful zero and hence allow relative comparison of two measurements (ie division of one measurement by the other)

The coefficient of variation is useful because the standard deviation of data must always be understood in the context of the mean of the data. In contrast, the actual value of the CV is independent of the unit in which the measurement has been taken, so it is a dimensionless number. For comparison between data sets with different units or widely different means, one should use the coefficient of variation instead of the standard deviation.

7. Standard Error

The standard error (SE) of a statistic (usually an estimate of a parameter) is the standard deviation of its sampling distribution.

The relationship between the standard error of the mean and the standard deviation is such that, for a given sample size, the standard error of the mean equals the standard deviation divided by the square root of the sample size. In other words, the standard error of the mean is a measure of the dispersion of sample means around the population mean.

Standard error, $\sigma_x = \frac{\sigma}{\sqrt{n}}$

Where,

σ is the standard deviation of the population.

n is the size (number of observations) of the sample.

8. Z-score (Standard score)

In statistics, the standard score is the number of standard deviations by which the value of a raw score (i.e., an observed value or data point) is above or below the mean value of what is being observed or measured. Raw scores above the mean have positive standard scores, while those below the mean have negative standard scores.

It is calculated by subtracting the population mean from an individual raw score and then dividing the difference by the population standard deviation. This process of converting a raw score into a standard score is called standardizing or normalizing.

If the population mean and population standard deviation are known, a raw score x is converted into a standard score by

$$Z = \frac{x - \mu}{\sigma}$$

where:

μ is the mean of the population.

σ is the standard deviation of the population.

9.Skewness

In probability theory and statistics, skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable about its mean. The skewness value can be positive, zero, negative, or undefined.

For a unimodal distribution, negative skew commonly indicates that the tail is on the left side of the distribution, and positive skew indicates that the tail is on the right. For example, a zero value means that the tails on both sides of the mean balance out overall; this is the case for a symmetric distribution, but can also be true for an asymmetric distribution where one tail is long and thin, and the other is short but fat.

Consider the two distributions in the figure just below. Within each graph, the values on the right side of the distribution taper differently from the values on the left side. These tapering sides are called tails, and they provide a visual means to determine which of the two kinds of skewness a distribution has:

negative skew: The left tail is longer; the mass of the distribution is concentrated on the right of the figure. The distribution is said to be left-skewed, left-tailed, or skewed to the left, despite the fact that the curve itself appears to be skewed or leaning to the right; left instead refers to the left tail being drawn out and, often, the mean being skewed to the left of a typical center of the data. A left-skewed distribution usually appears as a right-leaning curve.

positive skew: The right tail is longer; the mass of the distribution is concentrated on the left of the figure. The distribution is said to be right-skewed, right-tailed, or skewed to the right, despite the fact that the curve itself appears to be skewed or leaning to the left; right instead refers to the right tail being drawn out and, often, the mean being skewed to the right of a typical center of the data. A right-skewed distribution usually appears as a left-leaning curve.

- If skewness is between −1 and −½ or between +½ and +1, the distribution is **moderately skewed**.
- If skewness is between −½ and +½, the distribution is **approximately symmetric**.

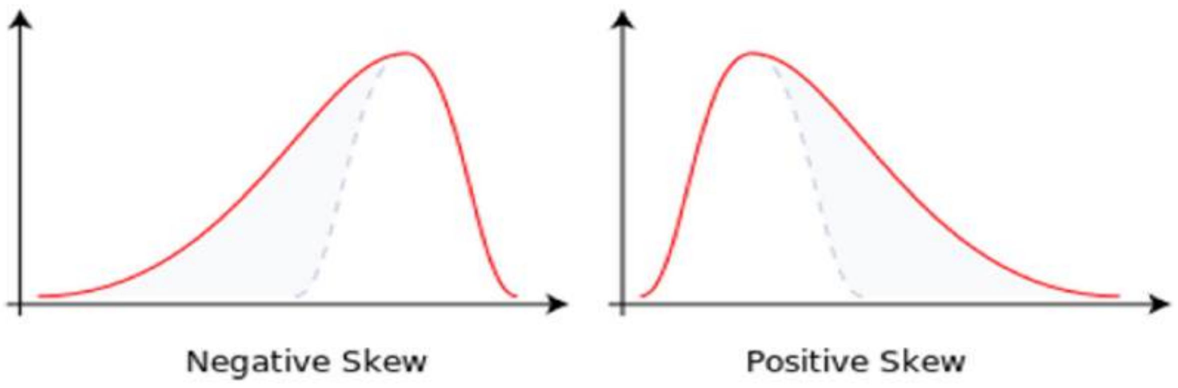


Fig 2.9.1: Negative and Positive Skewness of curve

skewness: $g_1 = m_3 / m_2^{3/2}$

where,

$m_3 = \sum(x-\bar{x})^3 / n$ and $m_2 = \sum(x-\bar{x})^2 / n$

$\bar{x}$ is the mean and n is the sample size, as usual. m_3 is called the **third moment** of the data set. m_2 is the **variance**, the square of the standard deviation.

The skewness can also be computed as g_1 = **the average value of z^3**, where z is the familiar z-score, $z = (x-\bar{x})/\sigma$

10. Kurtosis

kurtosis is a measure of the "tailedness" of the probability distribution of a real-valued random variable.

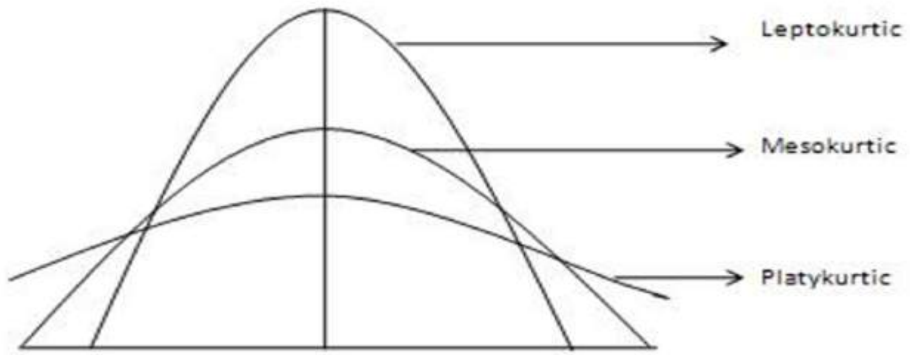


Fig 2.10.1: Graph of various Kurtosis

kurtosis: $a_4 = m_4 / m_2^2$

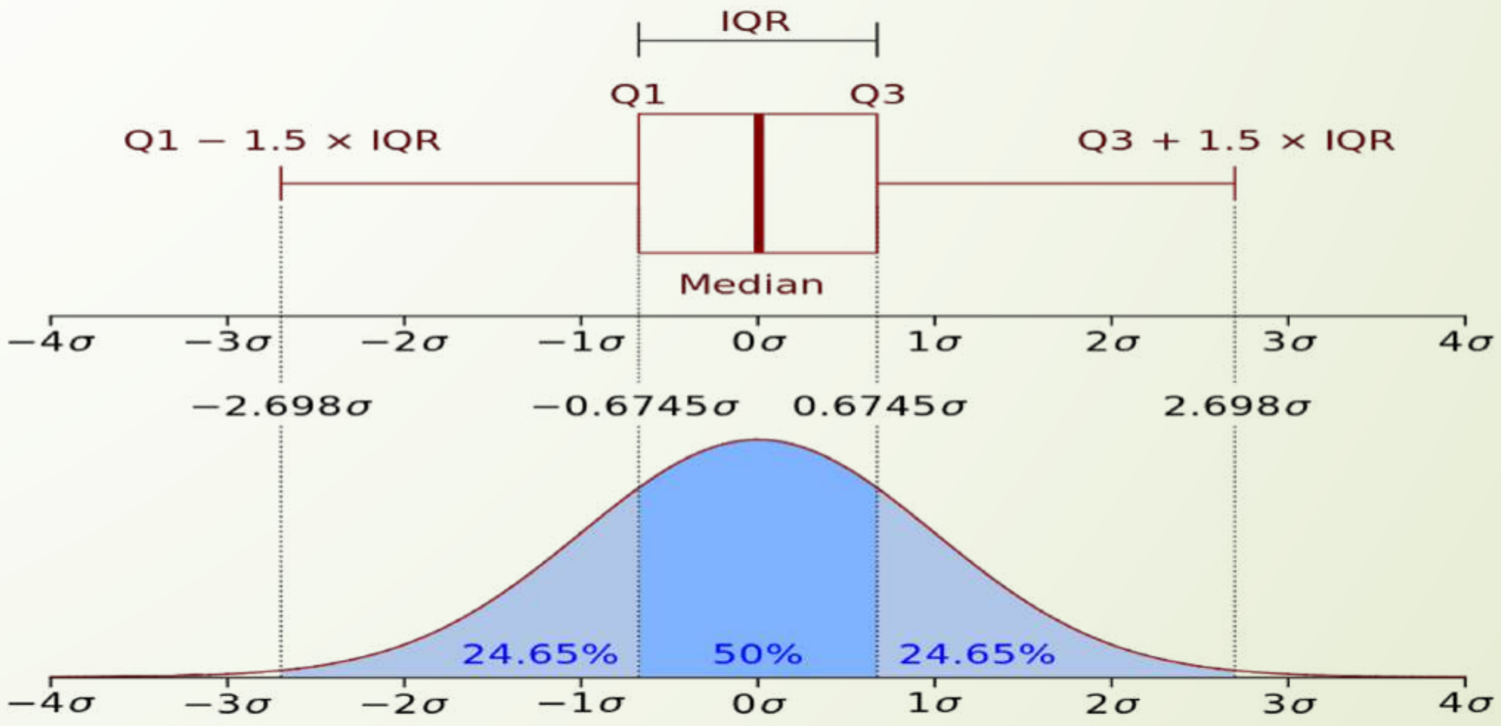
where

$m_4 = \sum(x-\bar{x})^4 / n$ and $m_2 = \sum(x-\bar{x})^2 / n$

$\bar{x}$ is the mean and n is the sample size, as usual. m_4 is called the **fourth moment** of the data set. m_2 is the **variance**, the square of the standard deviation.

11. Quartile

A quartile is a type of quantile which divides the number of data points into four more or less equal parts, or quarters. The first quartile (Q1) is defined as the middle number between the smallest number and the median of the data set. It is also known as the lower quartile or the 25th empirical quartile and it marks where 25% of the data is below or to the left of it (if data is ordered on a timeline from smallest to largest). The second quartile (Q2) is the median of a data set and 50% of the data lies below this point. The third quartile (Q3) is the middle value between the median and the highest value of the data set. It is also known as the upper quartile or the 75th empirical quartile and 75% of the data lies below this point.



12 Correlation coefficient

The correlation coefficient (ρ) is a measure that determines the degree to which the movement of two variables is associated. The most common correlation coefficient, generated by the Pearson product-moment correlation, may be used to measure the linear relationship between two variables. However, in a non-linear relationship, this correlation coefficient may not always be a suitable measure of dependence.

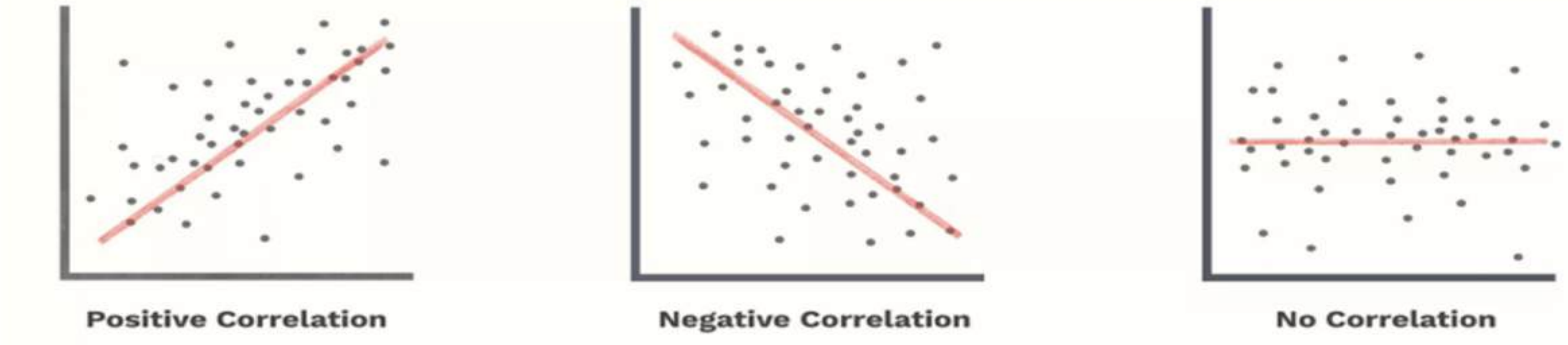


Fig 3.1.1: Types of correlation

$$correlation\ coefficient, r = \frac{cov(x,y)}{\sigma_x * \sigma_y}$$

Where,
Numerator is covariance of two signal x and y
Denominator is product of standard deviation of two signal.

13. Correlation between signals

In statistics, correlation or dependence is any statistical relationship, whether causal or not, between two random variables. In the broadest sense correlation is any statistical association, though it commonly refers to the degree to which a pair of variables are linearly related.

$$r_{xy} = \frac{\sum_{i=1}^{N}(x_i - \mu_x) * (y_i - \mu_y)}{S_x * S_y}$$

Where

μ_x is mean of x_i

μ_y is mean of y_i

S_x is standard deviation of x_i

S_y is standard deviation of y_i

We used xcorr function to find cross correlation between two signals

Syntax of xcorr function is,

Xcorr(x, y, 'normalized option')

Normalization option, specified as one of the following.

'none' — Raw, unscaled cross-correlation. 'none' is the only valid option when x and y have different lengths.

'biased' — Biased estimate of the cross-correlation:

$$R_{xy,biased} = \frac{1}{N} * R_{xy}(m)$$

'unbiased' — Unbiased estimate of the cross-correlation:

$$R_{xy,unbiased} = \frac{1}{N - |m|} * R_{xy}(m)$$

'normalized' or 'coeff' — Normalizes the sequence so that the autocorrelations at zero lag equal 1:

$$R_{xy,coeff} = \frac{1}{\sqrt{R_{xx}(0) * R_{yy}(0)}} * R_{xy}(m)$$

14 Power Spectral Density

The power spectrum S (f) of a time series x (t) describes the distribution of power into frequency components composing that signal. According to Fourier analysis, any physical signal can be decomposed into a number of discrete frequencies, or a spectrum of frequencies over a continuous range. The statistical average of a certain signal or sort of signal (including noise) as analyzed in terms of its frequency content, is called its spectrum.

The power spectral density (PSD) then refers to the spectral energy distribution that would be found per unit time, since the total energy of such a signal over all time would generally be infinite. Summation or integration of the spectral components yields the total power.

Power spectral density function is a very useful tool if you want to identify oscillatory signals in your time series data and want to know their amplitude. Power spectral density tells us at which frequency ranges variations are strong and that might be quite useful for further analysis.

the power spectral density is simply defined as the Fourier transform of autocorrelation.

FFT provides us spectrum density(i.e. frequency) of the time-domain signal. So, PSD is defined taking square the of absolute value of FFT.

Then a single estimate of the PSD can be obtained through summation rather than integration:

$$x(w) = \sum_{n=1}^{N} x(n) * e^{-i\omega n\Delta t}$$

$$S_{xx}(w) = \frac{(\Delta t)^2}{T} * abs\left(\sum_{n=1}^{N} x(n) * e^{-i\omega n\Delta t}\right)^2$$

Chapter 4

Results and Analysis

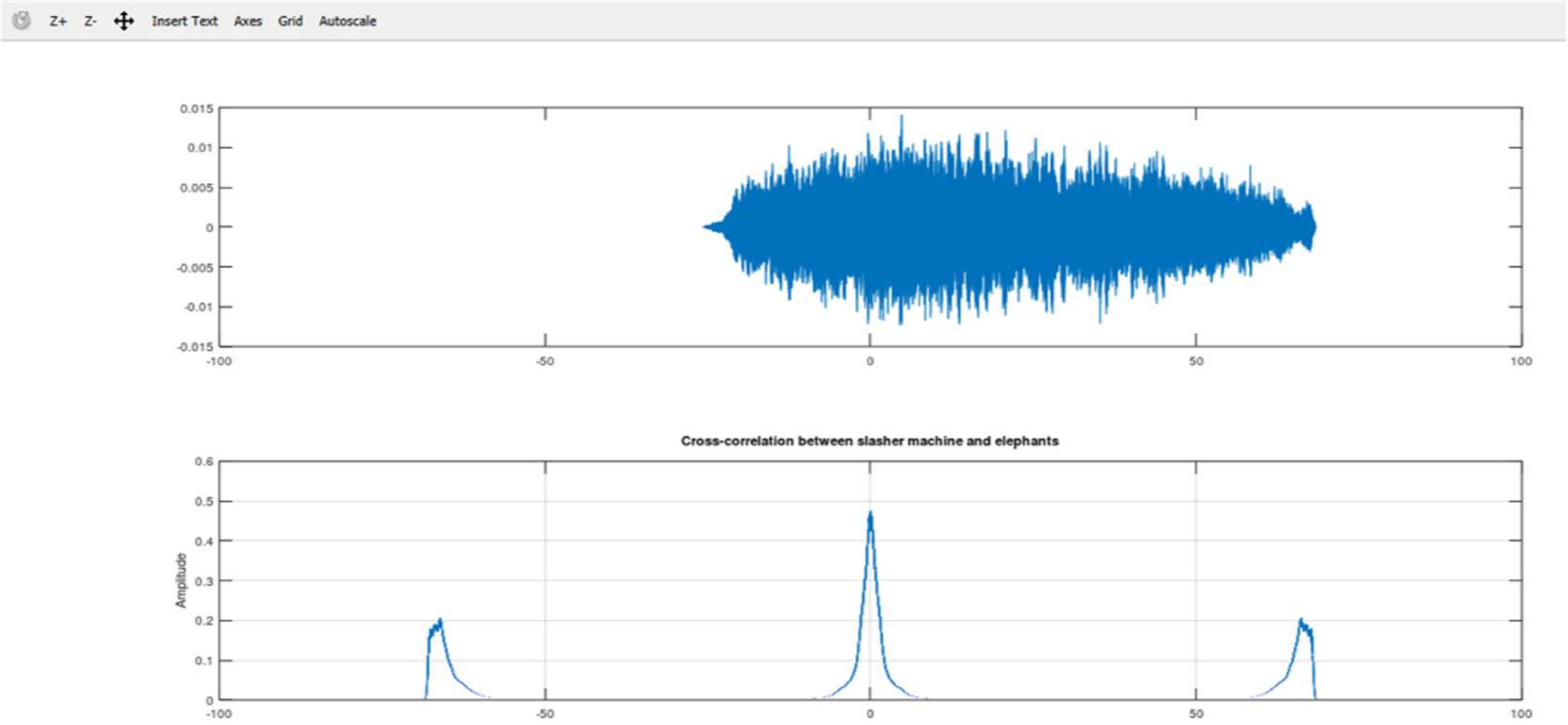


Figure 4.1: correlation between slasher machine and elephant

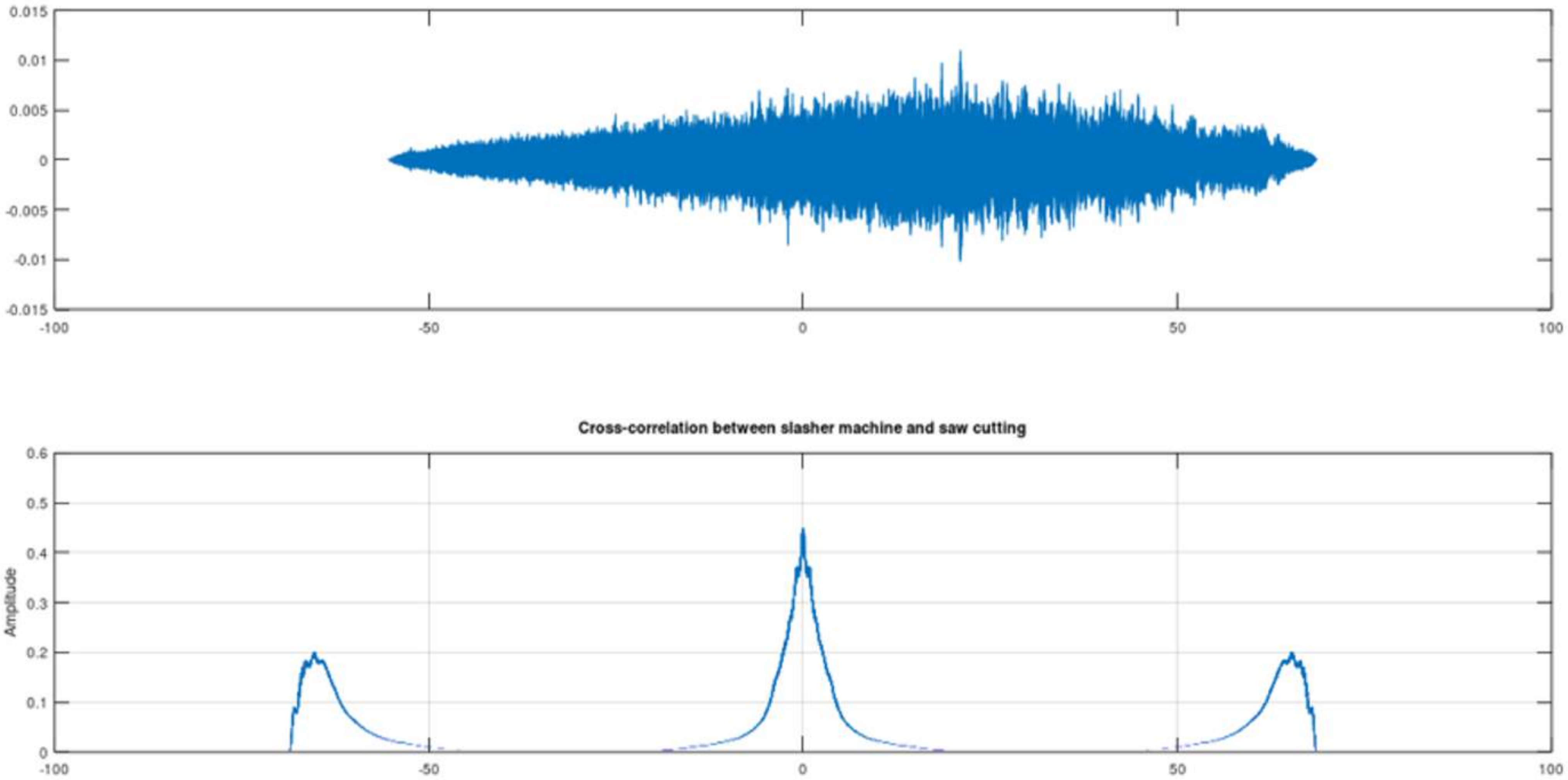


Figure 4.2: correlation between slasher machine and saw cutting

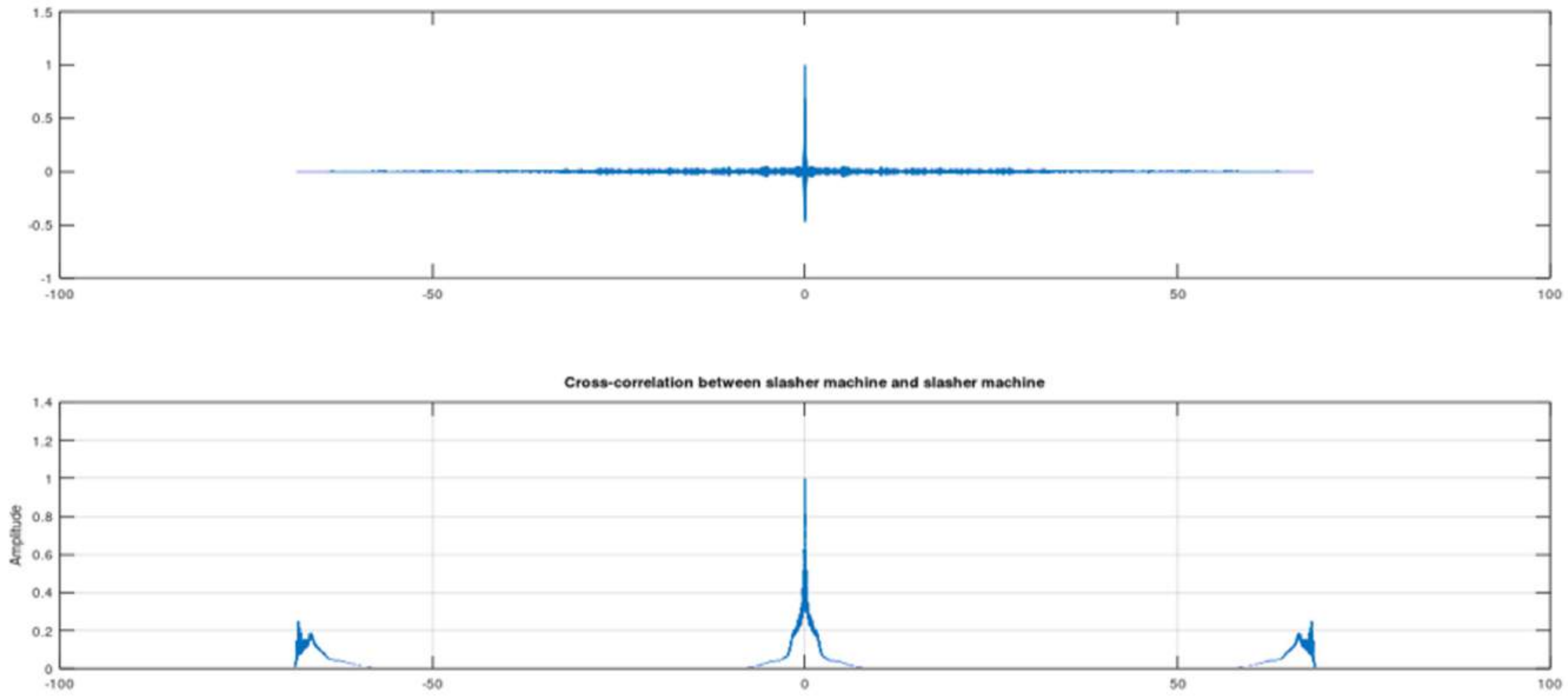


Figure 4.3: correlation between slasher machine and slasher machine

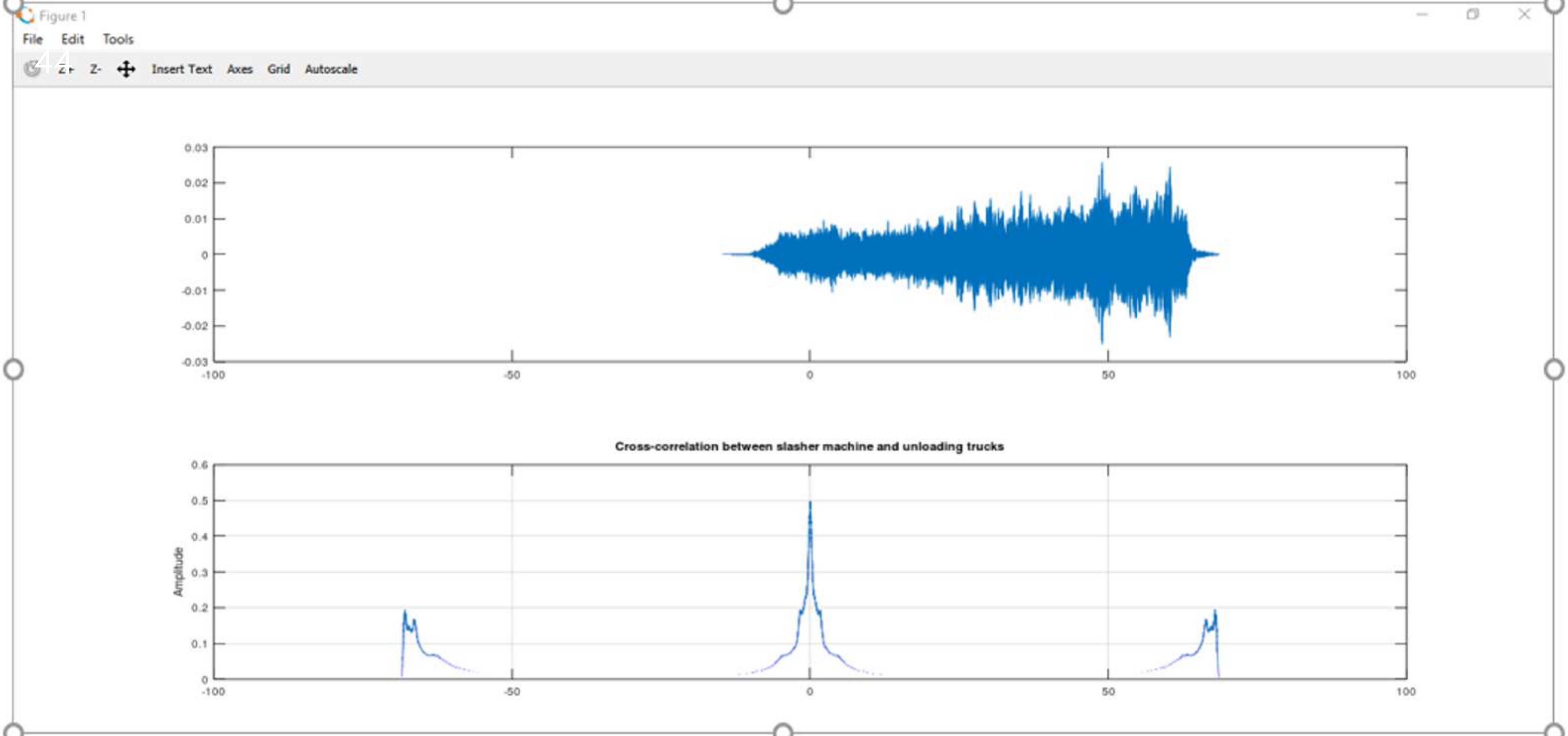


Figure 4.4: correlation between slasher machine and unloading trucks

Table 4.1: simulation result of correlation between slasher machine and another signal

Slasher with different signal	Correlation coefficient	Max of normalized correlation from xcorr	Time lag(s)
Axe _chopping	0.0011512	0.013017	54.116
Elephants	0.00098402	0.014112	4.775
Loading on hydraulic trucks	0.0021096	0.019063	13.710
Saw cutting	-0.00064524	0.011002	21.503
Skidder machine	-0.0022272	0.008950	26.769
Slasher machine	1	1	0
Tree breaking	0.00077336	0.006650	63.522
Unloading trucks	0.0024763	0.025797	48.959

We can see clearly from table only for slasher machine the correlation coefficient is 1 because it is compared with itself. If signal correlated, we will get maxima at zero time instant as shown in figure above.

Now noise analysis is also added to see how correlation coefficient varies with addition of white noise or random noise using random noise function.

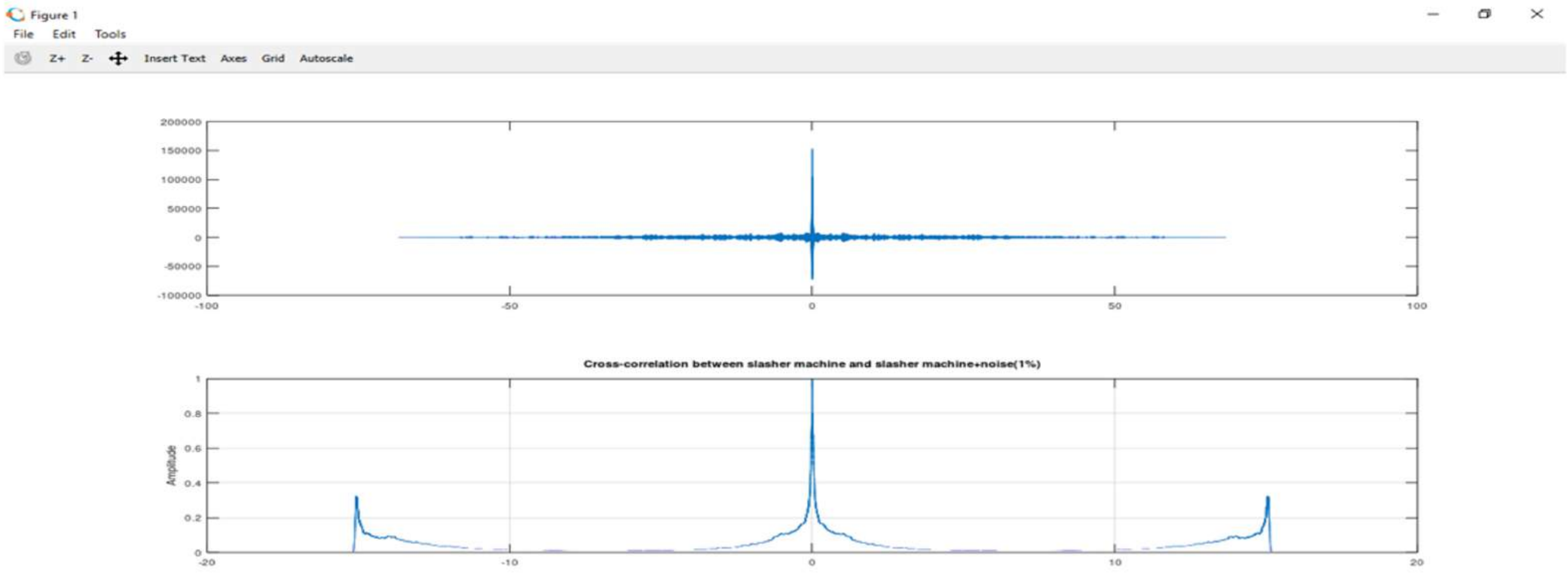


Figure 4.5: slasher machine signal is compared with slasher machine signal + 1% noise

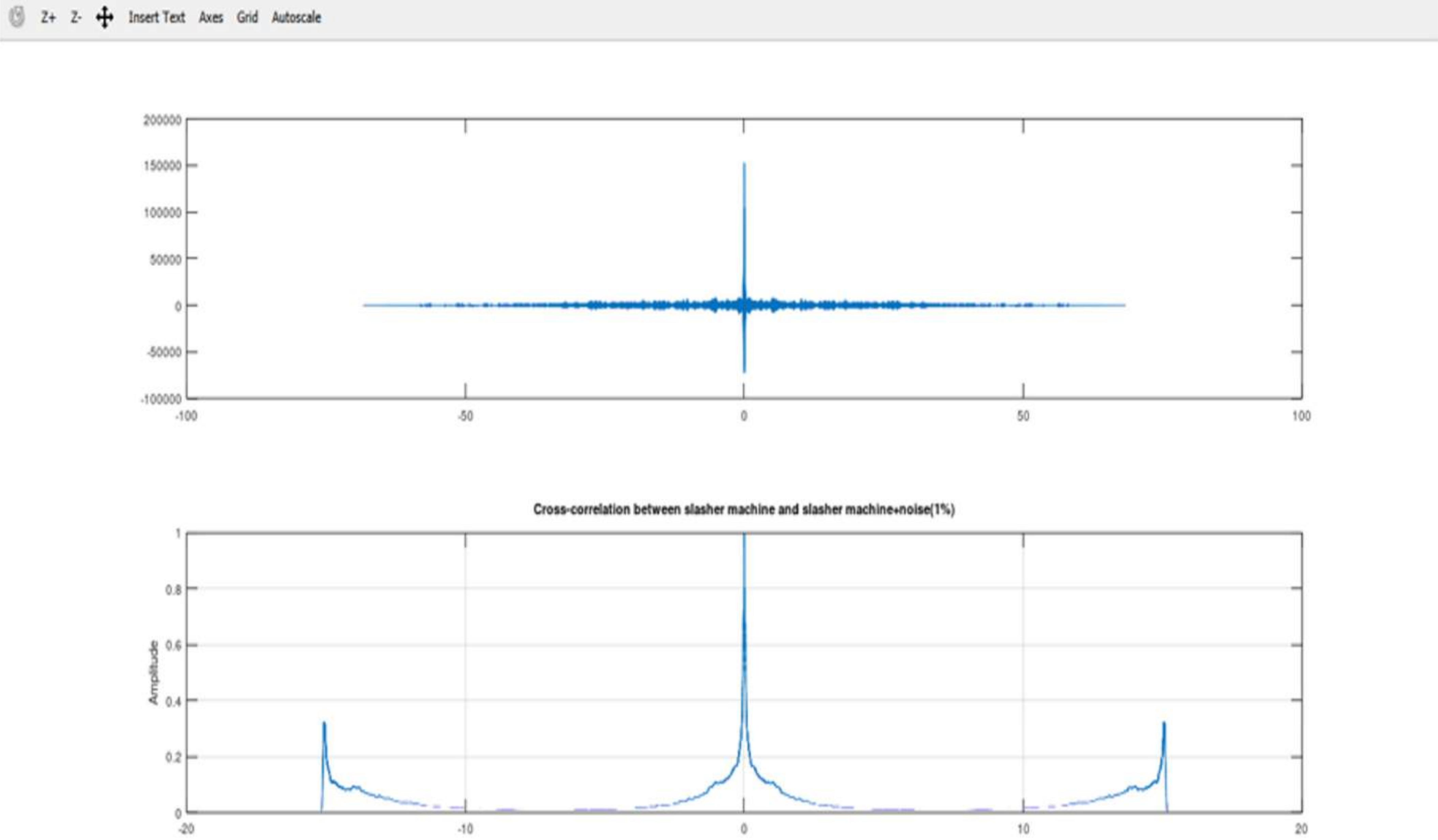


Figure 4.6: slasher machine signal is compared with slasher machine signal+ 5% noise

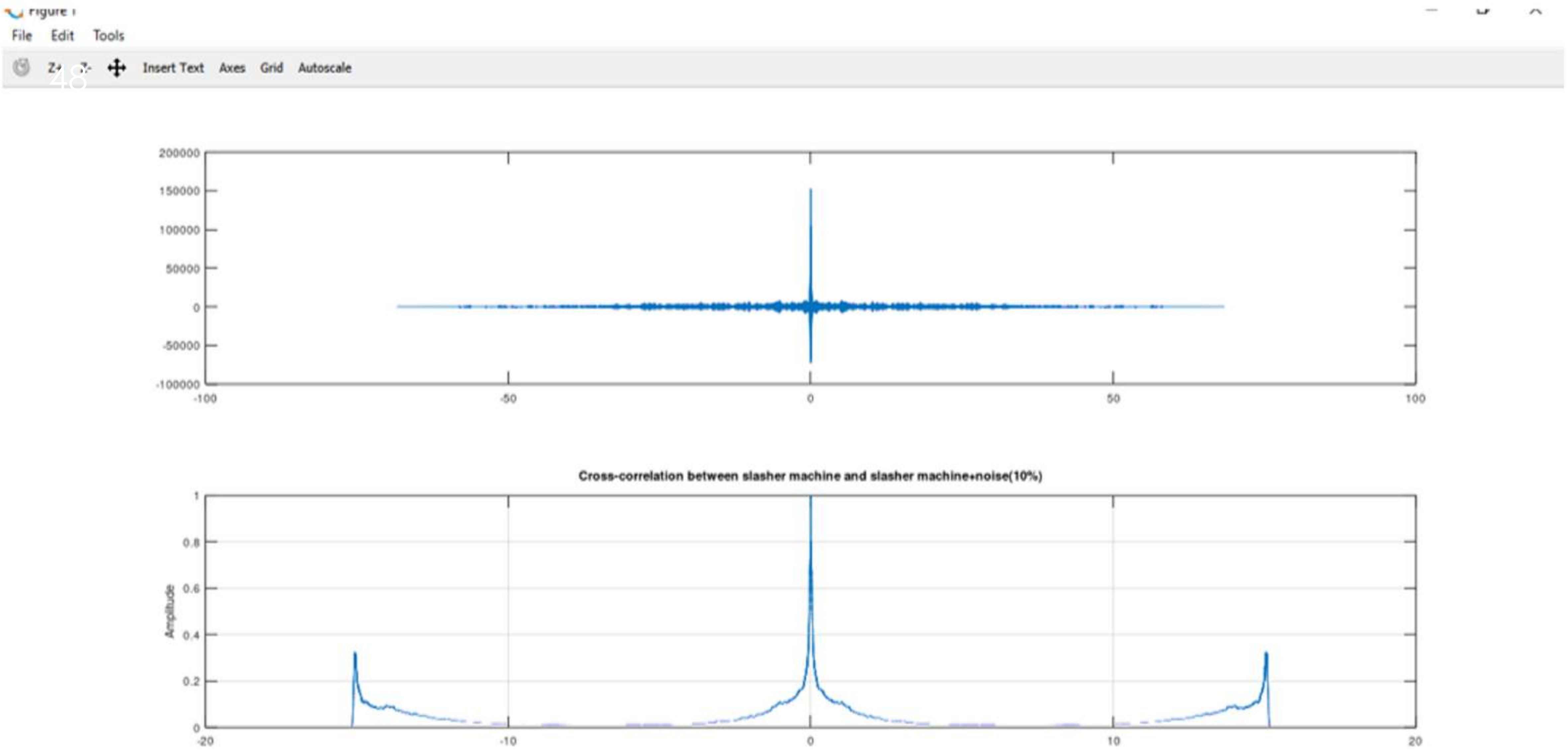


Figure 4.7: slasher machine signal compared with slasher machine signal + 10% noise

Below tables gives numerical result of correlation of various signal and shows how correlation coefficient decreases with addition of noise to the signal

Table 4.2: signal is compared with addition of 10 percentage of noise

Signal with signal + noise (10%)	Correlation coefficient	Max of correlation from xcorr	Time lag
Axe _chopping	0.44730	277.356698	0
Elephants	0.75815	17836.14328	0
Loading on hydraulic trucks	0.71264	40686.63220	0
Saw cutting	0.81836	49486.92584	0
Skidder machine	0.74757	22178.083802	0
Slasher machine	0.91351	152812.10167	0
Tree breaking	0.51163	974.505458	0
Unloading trucks	0.79970	11870.48629	0

Table 4.3: signal comparison with addition of 5 percentage noise

Signal with signal + noise (5%)	Correlation coefficient	Max of correlation from xcorr	Time lag
Axe _chopping	0.70444	275.953659	0
Elephants	0.91885	17844.232633	0
Loading on hydraulic trucks	0.89735	40698.156850	0
Saw cutting	0.94348	49510.096485	0
Skidder machine	0.91375	22158.849005	0
Slasher machine	0.97608	152823.454402	0
Tree breaking	0.76639	974.443829	0
Unloading trucks	0.93607	11860.61308	0

Table 4.4: signal comparison with addition of 1 percentage noise

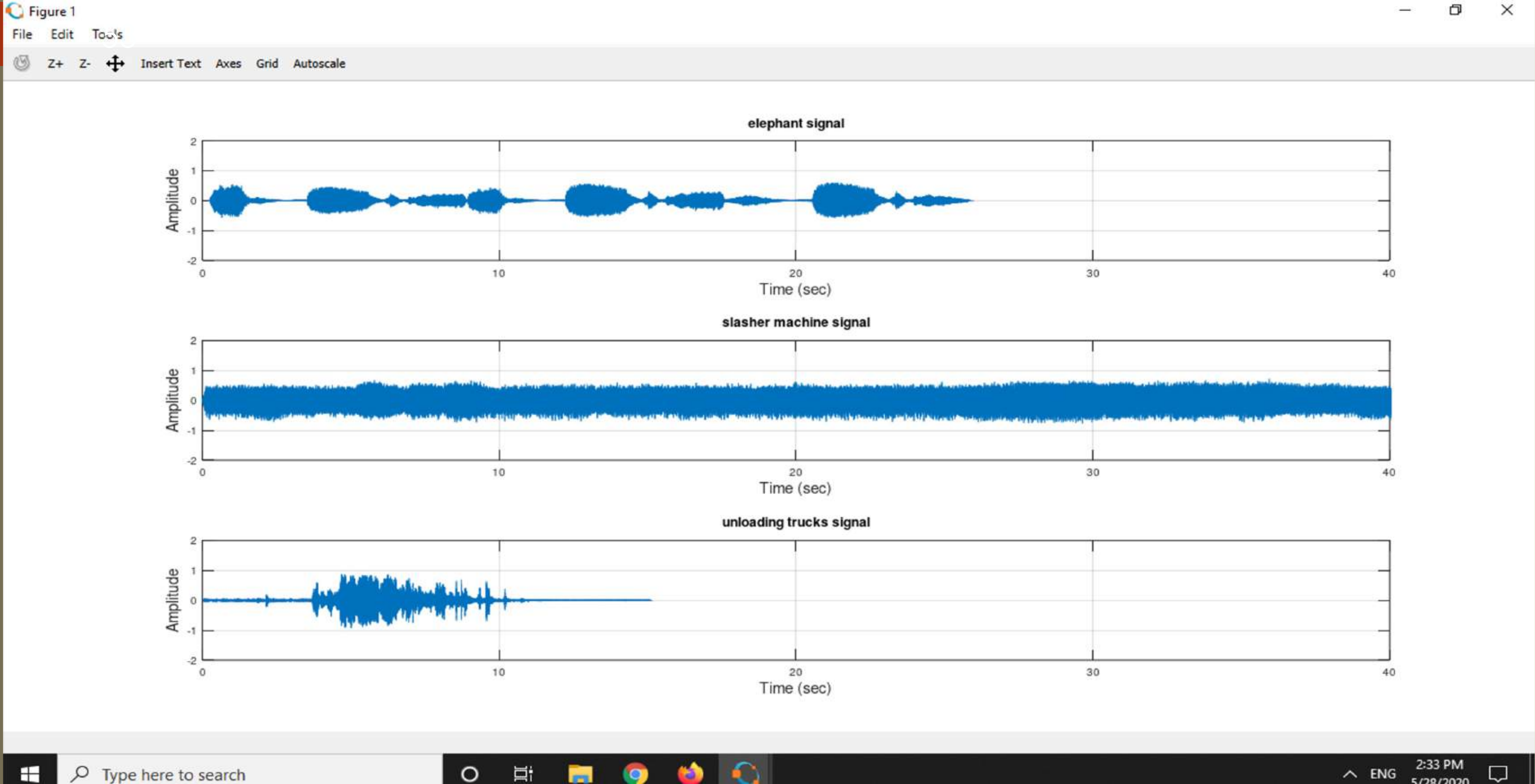
Signal with signal + noise (1%)	Correlation coefficient	Max of correlation from xcorr	Time lag
Axe _chopping	0.98035	276.771890	0
Elephants	0.99633	17829.8981	0
Loading on hydraulic trucks	0.99520	40688.550830	0
Saw cutting	0.99754	49511.569004	0
Skidder machine	0.99607	22156.84657	0
Slasher machine	0.99901	152817.6573	0
Tree breaking	0.98618	974.398337	0
Unloading trucks	0.99718	11859.84856	0

Table 4.5 : statistical parameters of various Signal

Parameter	Axe chopping	Elephants	Loading on hydraulic trucks	Saw cutting	Skidder machine	Slasher machine	Tree breaking	Unloading trucks
Energy	276.42	17828.73	40688.995	49515.33	22155.14	152815.33	974.20	11861.0367
Zero crossing	26292	42713	100591	195430	66222	92469	23650	32923
mean	0.000042	0.000003	0.000011	0.000004	0.000011	0.000006	0.001462	-0.000003
Standard deviation	0.049788	0.067294	0.101662	0.142300	0.112417	0.224506	0.059530	0.132950
Avg. power	0.002479	0.004529	0.010335	0.020249	0.012638	0.050403	0.003546	0.017676
Avg. magnitude	0.010613	0.023040	0.076969	0.110095	0.084125	0.178751	0.032371	0.063774
range	1.784973	1.194031	1.839630	1.785767	1.672638	1.719055	1.350891	1.812592
Coefficient of variance	1197.1164	21113.69	9300.08	32481.25	9826.15	36371.11	40.7283	-52167.760
Standard error	0.000149	0.000059	0.000051	0.000091	0.000085	0.000129	0.000114	0.000162
z-score (maxima)	19.228745	8.962788	9.145998	5.927407	7.400119	3.843046	11.3141	6.614035
z-score (minima)	-16.62291	-8.780579	-8.949590	-6.621888	-7.478777	-3.814012	-11.3786	-7.019596

Skewness	0.87216	0.060891	0.010489	0.0018442	-0.077488	-0.075738	-0.015455	-0.0048615
Kurtosis	98.402	6.7352	5.1331	3.8258	4.7853	2.8736	17.079	12.659
Lower quartile	-0.000305	0.000000	-0.060547	-0.088959	-0.063721	-0.148499	-0.01257	-0.015778
Upper quartile	0.000366	0.000000	0.060272	0.088928	0.064911	0.149994	0.014130	0.015808
Interquartile range	0.000671	0.000000	0.120819	0.177887	0.128632	0.298492	0.026703	0.031586

Power Spectrum Density



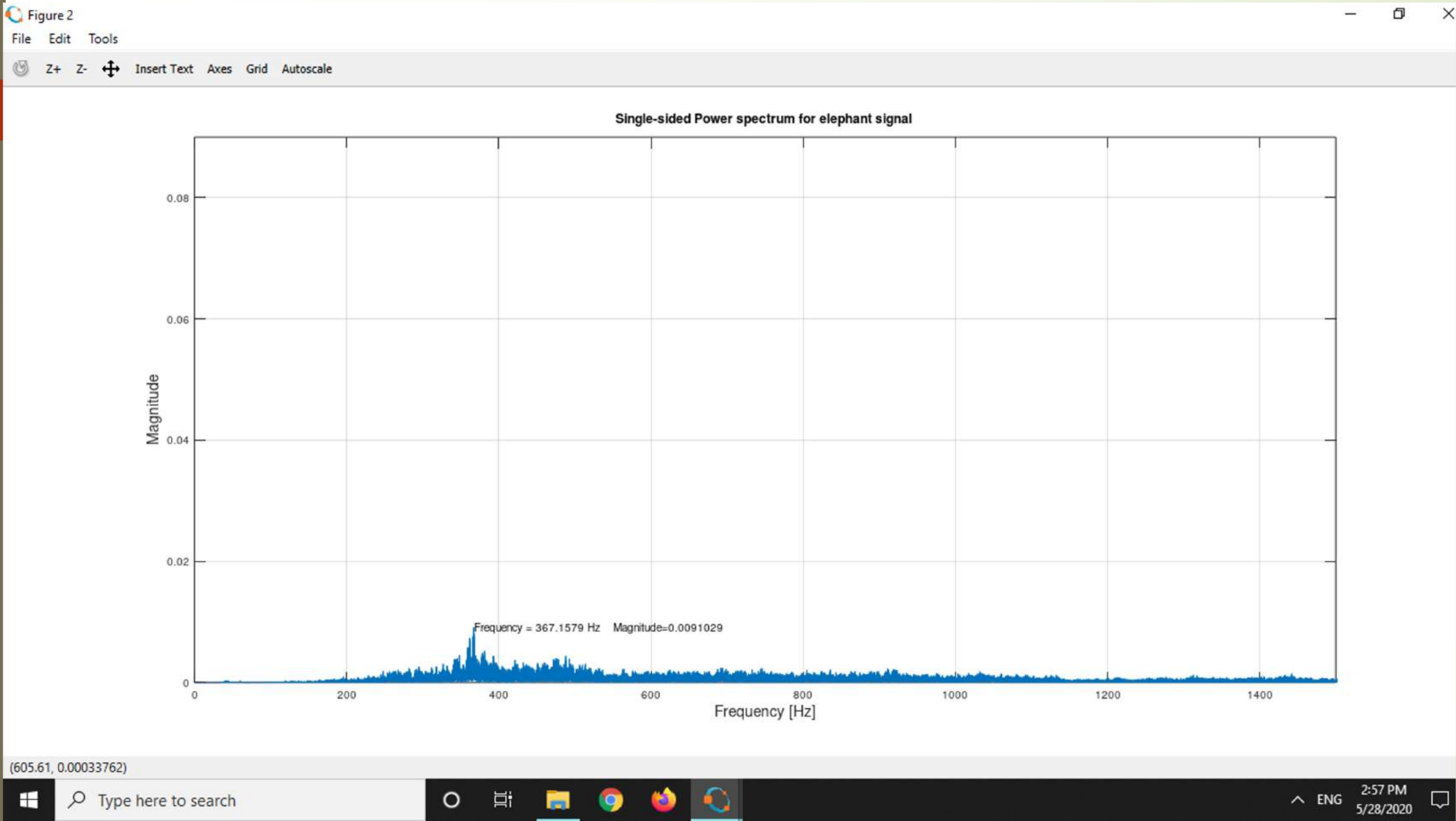


Figure 2
File Edit Tools
Z+ Z- Insert Text Axes Grid Autoscale
Single-sided Power spectrum for elephant signal
Magnitude
0.08
0.06
0.04
0.02
0
Frequency = 367.1579 Hz Magnitude=0.0091029
0
200
400
600
800
1000
1200
1400
Frequency [Hz]
(605.61, 0.00033762)
Type here to search
ENG
2:57 PM
5/28/2020

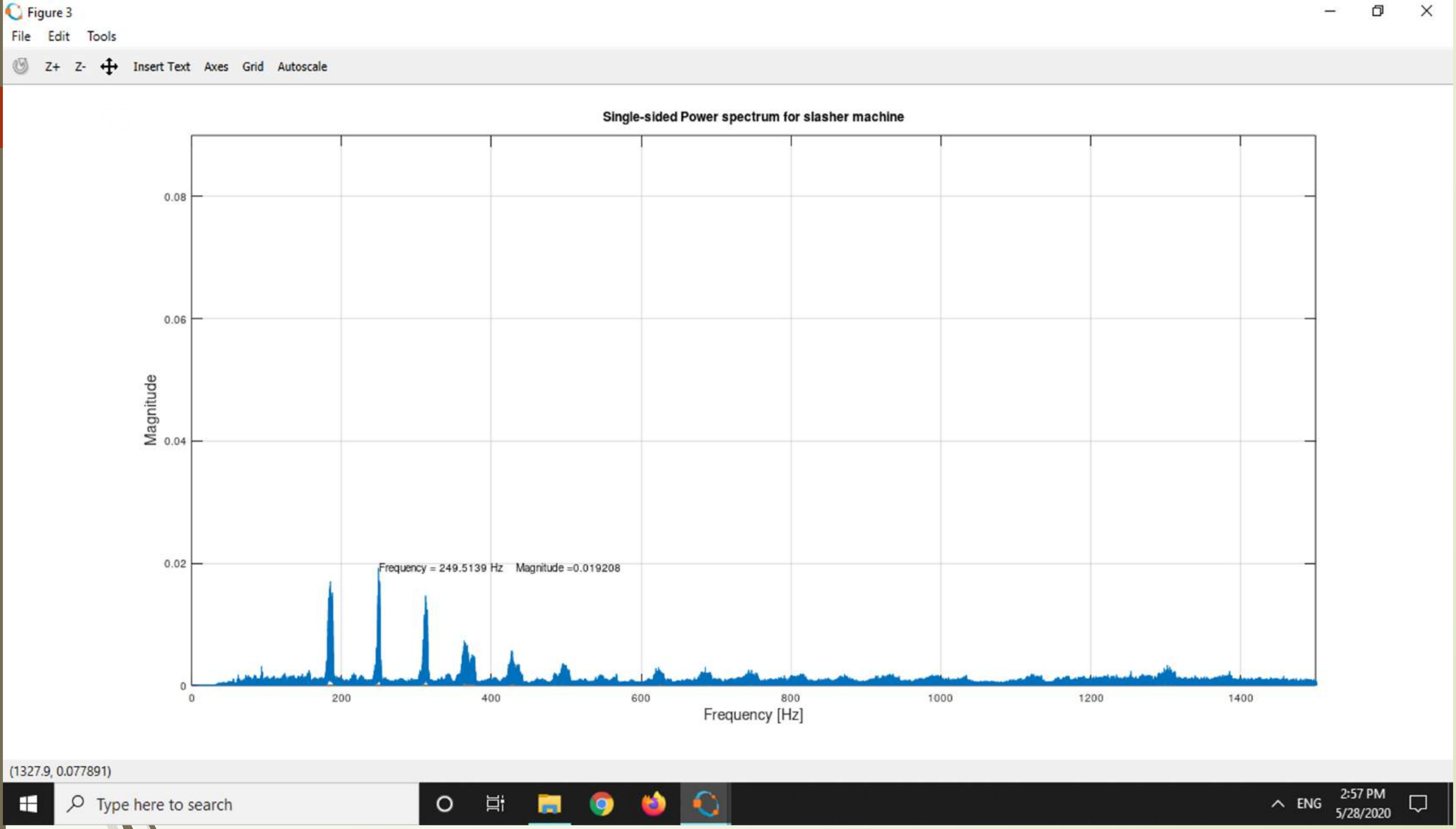


Figure 3
File Edit Tools
Z+ Z- Insert Text Axes Grid Autoscale
Single-sided Power spectrum for slasher machine
Magnitude
0.08
0.06
0.04
0.02
0
Frequency = 249.5139 Hz Magnitude =0.019208
0
200
400
600
800
1000
1200
1400
Frequency [Hz]
(1327.9, 0.077891)
Type here to search
ENG
2:57 PM
5/28/2020

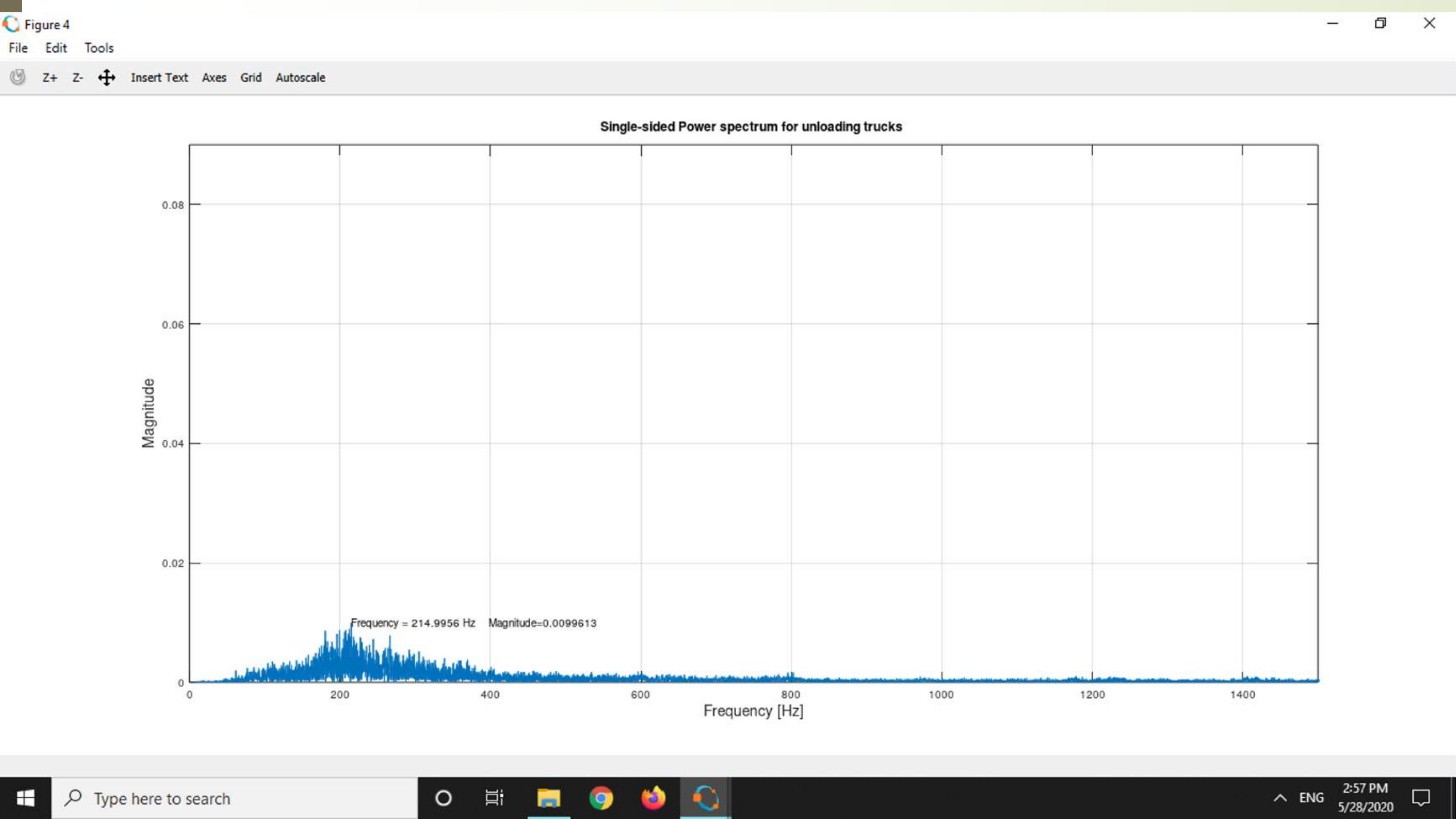


Figure 4
File Edit Tools
Z+ Z- Insert Text Axes Grid Autoscale
Single-sided Power spectrum for unloading trucks
Magnitude
0.08
0.06
0.04
0.02
0
Frequency = 214.9956 Hz Magnitude=0.0099613
0
200
400
600
800
1000
1200
1400
Frequency [Hz]
Type here to search
ENG
2:57 PM
5/28/2020

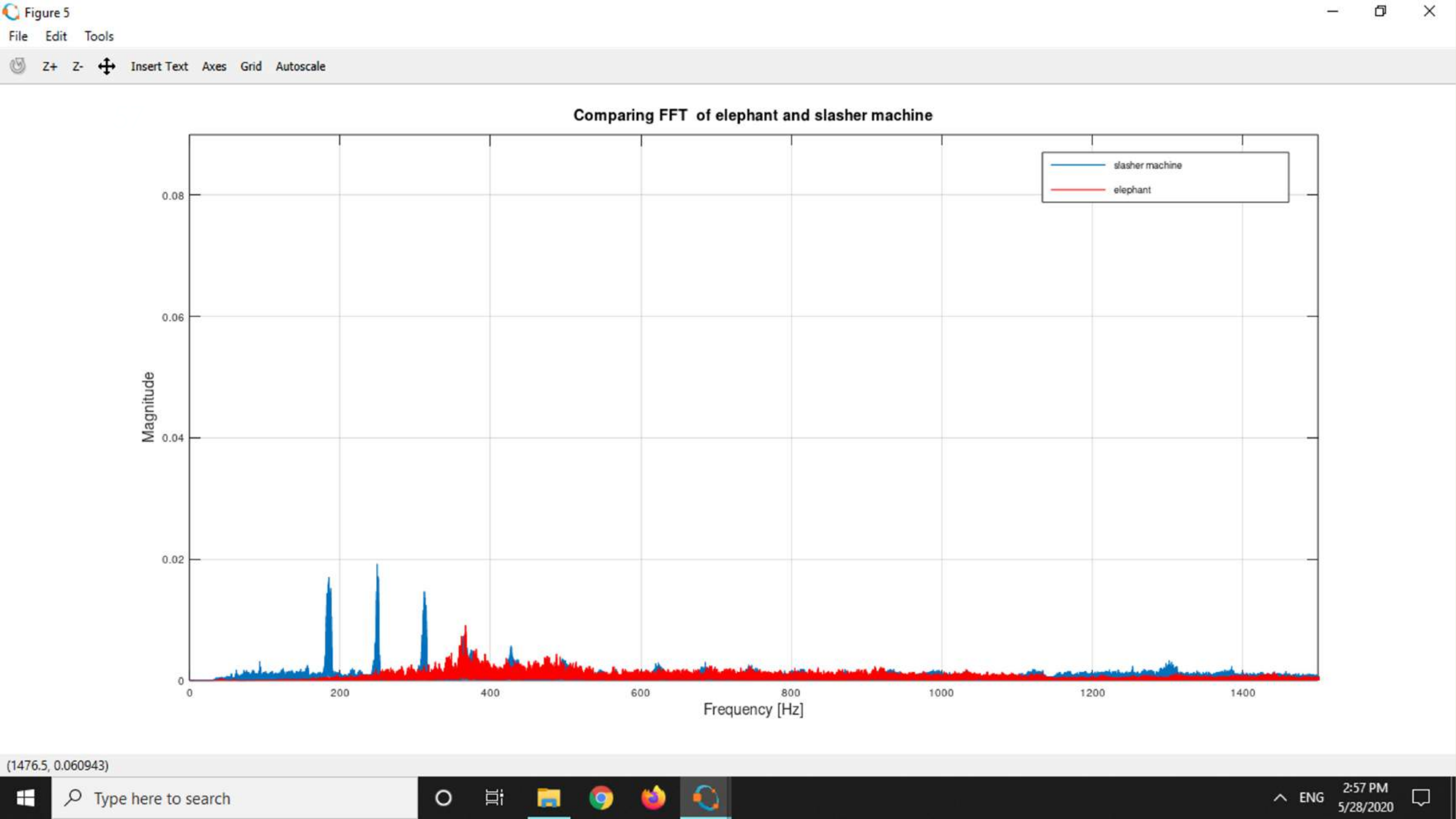


Figure 5
File Edit Tools
Z+ Z- Insert Text Axes Grid Autoscale
Comparing FFT of elephant and slasher machine
slasher machine
elephant
Magnitude
0.08
0.06
0.04
0.02
0
0
200
400
600
800
1000
1200
1400
Frequency [Hz]
(1476.5, 0.060943)
Type here to search
ENG
2:57 PM
5/28/2020

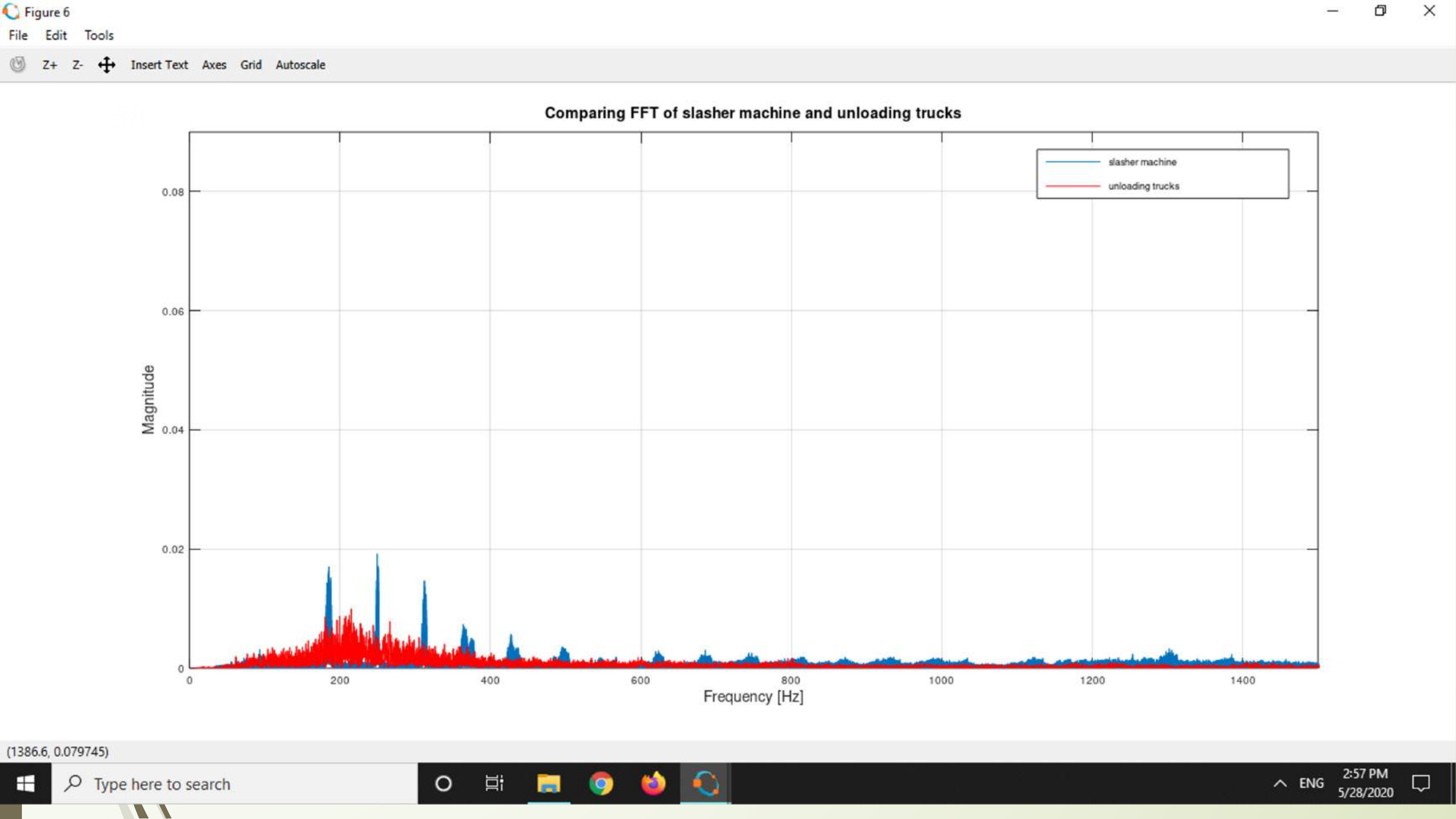


Figure 6
File
Edit
Tools
Z+
Z-
Insert Text
Axes
Grid
Autoscale
Comparing FFT of slasher machine and unloading trucks
slasher machine
unloading trucks
Magnitude
0.08
0.06
0.04
0.02
0
0
200
400
600
800
1000
1200
1400
Frequency [Hz]
(1386.6, 0.079745)
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ENG
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5/28/2020

Chapter 5

Conclusion and Future Scope

Proposed Conclusions

- Our ultimate goal is to extract the necessary data from nodes using sensor network which is to be compared with pre fed data in the central server to obtain the essential results
- One of the most important constraints in wireless sensor networks is the energy consumption. To decrease the power consumption, the number of transmitted messages in network is decreased.
- Range of transmission can be increased by using more powerful transmitters and use of convenient mesh networks
- Various techniques for frequency transformation and analysis of signals can be implemented to find the most accurate and precise algorithm.

Applications

- Efficient and secure data abstraction over a mesh network
- Prevent forest fires through sensor data
- Alert over forest logging

Limitations

- Setting of nodes in forest
- Chances of time lag.

Future Scope

- 1. We have considered using MQTT protocol to send the data from the nodes in case of internet connected network. MQTT is a lightweight and faster protocol than HTTP hence preferable
- 2. The design process must consider all power penalties associated with signal degradation process.
- 3. The non-linearities that arise when noise is added to the channel must be minimized with increased efficiency in SNR ratio for more accurate signal detection and processing.

Future Scope

- 1. We have considered using MQTT protocol to send the data from the nodes in case of internet connected network. MQTT is a lightweight and faster protocol than HTTP hence preferable
- 2. The design process must consider all power penalties associated with signal degradation process.
- 3. The non linearities that arise when noise is added to the channel must be minimized with increased efficiency in SNR ratio for more accurate signal detection and processing.
- 4. Deep learning algorithm can be used instead of FFT to increase the accuracy of data.

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THANK YOU